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Data-Driven Methods in Spectroscopy

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Keywords

data analysis, computational methods, stellar surveys, stellar properties, stellar abundances

Abstract

Understanding the formation and evolution of the Milky Way and its constituents relies on making measurements of data that are faithful representations of the underlying physical system. Data-driven spectral models function in several capacities in this context; generating spectra, extracting encoded information, and connecting that information to theory. We review how data-driven methods have changed the spectroscopic landscape and assess their future:

- A core utility of data-driven models is that they deliver precise stellar measurements at scale and with modest computational cost, providing consistent parameters and abundances across surveys, including for low-resolution and low signal-to-noise spectra. This circumvents prior limitations and unlocks the statistical power of large datasets.
- Entirely new avenues of study have been opened up by inferred spectroscopic ages, distances, and evolutionary states, whereas residual analyses have uncovered nonstellar signals and allow a novel discovery space.
- Biases can be introduced or inherited by data-driven models; identifying and accounting for them is important for accurate interpretation.
- Data-driven models do not replace theoretical models; together, they enable the interpretation and parameterization of spectra.
- A key benefit of data-driven frameworks, alongside public releases of survey data, is democratization of spectroscopic analysis.
- Future facilities and computational tools will expand data-driven methods and their utility. Although this review focuses on the stellar temperature range of 3,000–7,000 K, the methods are general.

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1. INTRODUCTION

Astronomy has entered the long-anticipated era in which *Gaia* has delivered photometry, astrometry, and parallaxes for billions of stars (Gaia Collab. et al. 2016, 2018, 2023). These data are complemented by hundreds of millions of stellar spectra from both satellite and ground-based facilities (e.g., Steinmetz et al. 2006, Gilmore et al. 2012, Zhao et al. 2012, De Silva et al. 2015, de Jong et al. 2016, Majewski et al. 2017, Kollmeier et al. 2025). The number of ground-based spectra—already in excess of 15 million—will double in the next decade. The orbital measurements combined with stellar parameters and element abundances of these stars provide the necessary information to reconstruct the history of the Milky Way Galaxy (e.g., Freeman & Bland-Hawthorn 2002, Rix & Bovy 2013, Bland-Hawthorn & Gerhard 2016, Deason & Belokurov 2024, Bonaca & Price-Whelan 2025).

This data-rich era has seen a corresponding growth in the necessary automation of processing, as well as in the extraction of information from these data (Jofré et al. 2019). New, automated data-driven approaches (i.e., whereby mathematical or computational models are learned directly from the data, often employing information from theoretical models) complement more traditional analysis methods that use ab initio physical stellar models (i.e., a theoretical stellar model constructed using model atmospheres, radiative transfer, atomic and molecular physics, abundance compositions, and spectrum synthesis) directly. Data-driven models of stars may use all or part of the spectra and/or photometry, or statistics of these, single stars with multiple observations, or populations of stars. In this review of data-driven methods in spectroscopy, we focus on models constructed using the spectra, but are inclusive of models that work with parameterizations or statistics of the spectra. Data-driven approaches are especially valuable in regimes in which physical models are uncertain or incomplete—as is often the case for stars—and they also serve as efficient computational tools for high-volume data analysis.

Ab initio physical stellar model: a theory-driven model, constructed from physical first principles without empirical calibration

Data-driven model: a mathematical or computational framework that learns relationships directly from observed data

Section 2 introduces the purpose and motivation behind data-driven modeling in spectroscopy. Although modern approaches use large datasets and rely on contemporary computational frameworks, data-based models, classifications that only use data, and automated measurement are not new in astronomy (e.g., Cannon & Pickering 1918, Deeming 1964, Kurtz 1983,

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von Hippel et al. 1994). A key historical example is early stellar classification; initial stellar classifications in the early twentieth century proceeded in advance of understanding of stellar physics and stellar atmospheres. Subsequent analysis of human-classified spectra in concert with theory led to transformational insights about the underlying physical process. This interplay between empirical and theoretical approaches continues today.

In Section 3, we examine the landscape of spectroscopic surveys, and how they connect to data-driven methods. Expanding computational capabilities have broadened the variety and utility of data-driven methods. In particular, the growth of data-driven tools within astronomy is occurring alongside substantial advances in artificial intelligence and machine learning, and development of open-source tools that enhance numerical analysis, optimization, and computational efficiency. Progress in the field is supported not only by the large volumes of data that have been observed but also by the diverse international teams who make both spectra and their parameterizations publicly available.

Section 4 delves into the ingredients of a data-driven model. There are now numerous methods that exist within the data-driven framework built for the purpose of extracting signals from spectra; labeling stellar spectra; performing calibration and residual analysis; and undertaking data-driven dynamics and nucleosynthesis. Data-driven models can be and have been built in separation from substantial consideration of their physical basis, and some data-driven modeling frameworks like instrument calibration are not intended to require any interpretative framework. However, for classification or assigning parameters and abundances to objects via label transfer, no physical meaning can be assigned to a system without a physical model. The interplay between data-driven modeling approaches and theory has not yet been pursued at scale in the present era and is highlighted as an opportunity for the coming decade (see the **Supplemental Text**).

Section 5 brings the survey landscape and contemporary opportunities together by examining modern data-driven methods and their applications. There are different choices in model architectures that support data-driven implementations. These choices encapsulate trade-offs among model simplicity and complexity, hierarchy, treatment of uncertainties, inclusion of prior or physical information, and whether the underlying data distribution itself is modeled or the model sets decision boundaries. These choices shape the speed, interpretability, precision, and parameter space over which a modeling framework is functional, as well as its ability to propagate and generalize to new data of heterogeneous quality. Different choices in modeling approaches are discussed and contrasted in Section 5.2. The areas in which data-driven models have opened new avenues of study and enabled new measurements are highlighted.

In Section 6, the limitations of data-driven models are examined. The sheer rate of data collection, speed of computational advances, and overall successes of data-driven approaches do not organically motivate intensive model interrogation and stress-testing. However, there may be important considerations revealed in adversarial analysis of data-driven frameworks, which is relevant to understanding the limitations of these modeling approaches.

The conclusion in Section 7 is that ultimately, data-driven modeling frameworks offer powerful, convenient, and effective solutions that can realize the joint-statistical power of large surveys and also extract information in the absence of physical models. They organically bridge scientific domains and open new avenues for scientific exploration. These methods will continue to evolve and presumably intersect with broader computational advances. As the pace of data analysis accelerates, careful and systematic validation of model outputs remains key. Ultimately, downstream inference from data-driven modeling frameworks is only as reliable as the extent to which these models faithfully capture the information encoded in the observed photons. A comparison of results from independent methods, many of which already exist, with the objective of identifying, quantifying, and understanding biases will underpin the constructive expansion of this realm.

Labeling: in spectra, these are numerical parameters that quantify the data, e.g., surface temperature and gravity, metallicity, individual abundances, turbulent velocities

Label transfer: using a reference set of labeled objects to build a model to infer labels for new objects

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2. PURPOSE AND MOTIVATION FOR A DATA-DRIVEN MODEL FOR SPECTROSCOPY

2.1. Where Data-Driven Models Are Employed

Data-driven models are useful in a number of contexts. These include separation of a stellar spectrum into constituent components, denoising, interpolation, extrapolation, and spectra generation, as well as feature discovery. One of the most popular use cases that underlies a growing literature of modeling efforts is labeling spectra with a set of numbers that describe its amplitude across wavelength, such as stellar parameters and element abundances.

Conventionally, a set of labels that describe stellar spectra of stars without a strong wind are delivered by comparisons and synthesis of theoretical stellar spectra. Often, this includes an iterative component and may include information from photometry or astrometry. For large survey data, a grid of theoretical spectra is typically first used to determine the best-fit parameters of the observed spectrum over four or more dimensions (i.e., $T_{\rm eff}$, $\log g$, [Fe/H],[1] [α/Fe], $v_{\rm mic}$, $v_{\rm macro}$, [C/Fe], [N/Fe]) (e.g., Smiljanic et al. 2014, García Pérez et al. 2016, Buder et al. 2025). Then, the analysis is restricted to windows around individual element absorption features, whereby the model abundance is varied to match the line profile of the observed to measure each [X/Fe]. Typically, there are substantial differences in the label scale for different surveys of the same stars (see **Figure 1**). These differences are inherited from different pipelines, simplified models, and inaccuracies in stellar physics (see the **Supplemental Text** and Jofré et al. 2019 for a review). Data-driven models, in concert with a set of reference stars whose labels faithfully represent the spectra, solve this problem of intersurvey label-scale differences. As long as there are stars in common between surveys, they enable the different surveys to be placed directly on the same label scale and the measurements from different surveys to be subsequently directly combined. The efficient and homogeneous transfer of labels between stellar surveys supports large-scale mapping efforts in deriving information from spectra for scientific endeavor. Using not only the outputs of models but also the residuals away from models can pinpoint anomalous populations and enable large-scale study of the material along the line of sight to the stars.

Data-driven models that enable label inference highlight the importance of high-fidelity[2] reference objects,[3] which serve as a ground truth and set the basis relationship between stellar flux and physical properties that are propagated at scale. With data-driven models built for label transfer, e.g., taking small high-fidelity stars in a survey (the reference objects) to construct a data-driven model to then use to label the full set of survey spectra, the inference is implicitly connected to ab initio models of spectra. In this capacity, the model that is trained using a reference set of spectra effectively propagates that label scale to new, consistently processed data, using the learned

[1] [X/Y] is the logarithmic abundance of element X relative to element Y in a star, compared to the same ratio in the Sun:

$$[\mathrm{X/Y}] = \log_{10}\left(\frac{N_{\rm X}}{N_{\rm Y}}\right)_{\rm star} - \log_{10}\left(\frac{N_{\rm X}}{N_{\rm Y}}\right)_{\odot}.$$

[2] Objects with precise labels and high-quality spectra, such as stars observed at high signal-to-noise ratios (SNRs), whose labels reliably represent the information encoded in their spectra. These serve as ideal reference objects for training models.

[3] The objects used to train the model, for which the labels are a faithful representation of the spectral flux (i.e., have precise and accurate labels). This includes observed spectra and/or photometry, for which the labels faithfully represent and generate the flux. Depending on the model framework and use case, reference data may consist of ensembles of stars or many observations of single stars. In the case of label transfer, the label space of the reference set should span that of the test.

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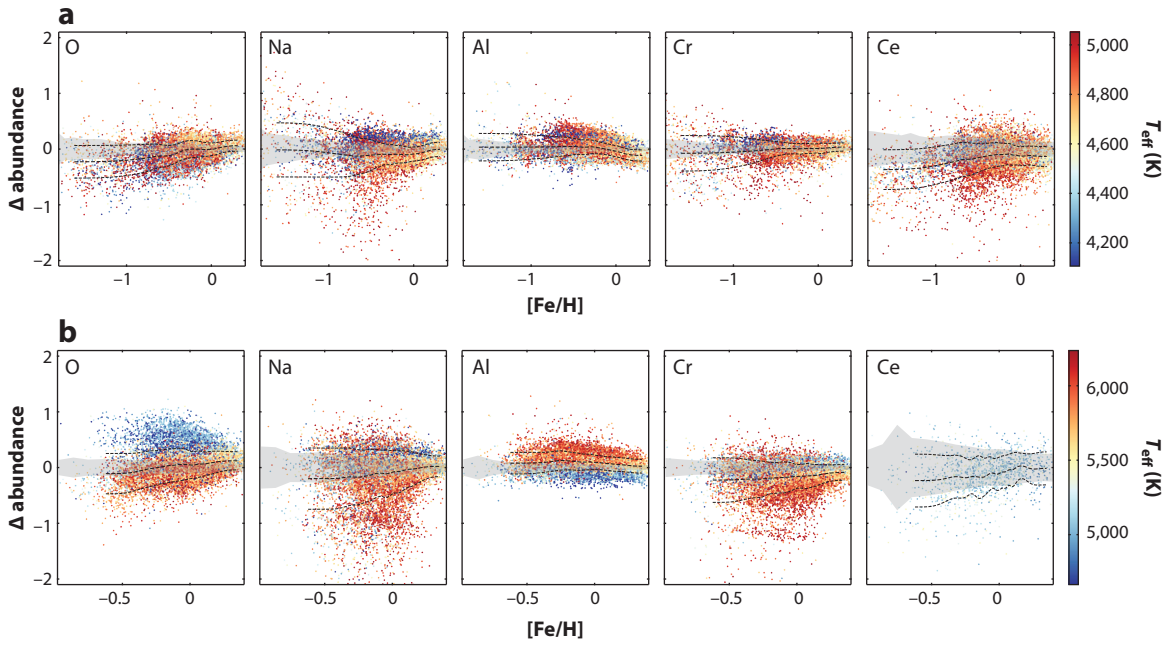


Figure 1

A comparison of five representative elements measured for the same stars in two different surveys. The Δ abundance (calculated as APOGEE DR17—GALAH DR4) is shown on the *y*-axis versus [Fe/H] on the *x*-axis, colored by T_{eff}, for ∼31,000 stars common to both surveys that meet the quality criteria set using recommended flags. Panel *a* shows ∼16,000 red giant stars, and panel *b* shows ∼15,000 main sequence stars. The gray shaded lines represent the scatter that is explained by the measurement uncertainty, and the dashed lines show the running mean and $1-\sigma$ standard deviation around the mean of the data. There are substantial deviations seen, in excess of the uncertainty, that show a temperature dependence for some elements. Abbreviations: APOGEE, Apache Point Observatory Galactic Evolution Experiment; GALAH, GALactic Archaeology with HERMES (High-Efficiency and Resolution Multi-Element Spectrograph).

relationship between spectral features and descriptive labels. Although to date there has been widespread use of data-driven modeling for precisely labeling stars in large surveys with stellar parameters, abundances, and ages using models built employing a set of reference objects with known labels, data-driven approaches are a utility to pursue and test ground truth or accuracy of theoretical models. It is possible to use the data to feed information back to physical models to ask where and how models and data diverge and why.

Notably, some data-driven frameworks obfuscate transparency, namely the connection between the information in the spectra and the labels. For label-inference applications in particular, it is beneficial to have an interpretative capability built in to understand how and why they produce the outputs they do. In other applications, it is not necessary to be able to interpret the model. In data processing, a goal is often to either separate signals of interest from nuisance signals or produce denoised, clean spectra to enable more precise measurements. These tasks rest on the assumption that observed spectra can be modeled as a combination of multiple contributing sources, including the stellar photosphere, circumstellar dust, interstellar medium, sky features, instrumental effects, and contamination from nearby astrophysical or artificial sources. A data-driven model can disentangle these components, provided sufficient training data exists to inform the variability associated with each source. This capability is particularly critical in high-precision domains such as exoplanet detection and elemental abundance analysis, where subtle spectral features must be

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separated from noise or systematic contaminants. Data-driven frameworks that operate in a data processing role do not necessarily require a high degree of traceability. Furthermore, the impact of instrumental artifacts and time-variable sky or photospheric signals cannot be readily analytically described.

The variety of data-driven methods that now exist reflects the different motivations behind them. Modeling choices often derive from distinct scientific goals and priorities, which shape how each approach is designed and applied. The numerous approaches that have been developed vary in simplicity, complexity, and flexibility. Models subsequently range in how interpretable they are and may or may not incorporate priors. Some approaches model the ensemble of information independently, and others consider it in a joint way. Data-driven frameworks also vary in the degree to which they can be scaled over parameter space and how robust or inclusive they can be of partial or missing data and noisy data. Different methods implement, either exclusively or in combination, generative model, discriminative model, deterministic model, or probabilistic model architectures. Regardless of the chosen approach, valuable intuition can be gained by directly exploring the data and comparing the outputs of different models applied to the same set of stars.

Priors: external knowledge incorporated into a model, typically as a probability distribution over parameters not derived from the data being modeled

Generative model: learns the full data distribution, i.e., the joint probability of the inputs and outputs, enabling synthesis

Discriminative model: learns decision boundaries so, e.g., predicts a star's parameters from its spectrum

Deterministic model: produces a single best-estimate prediction without representing uncertainty, e.g., returning the most likely $T_{\rm eff}$ without uncertainty estimates

Probabilistic model: outputs a full probability distribution over possible outcomes, e.g., providing the confidence distribution for all stellar parameters and abundances

2.2. Historical Perspectives on Data-Driven Models

At the beginning of the twentieth century, the physics of stars was not understood. However, this did not prohibit the classification of the spectra, which was done by eye and without the guidance of any physical model. In the early 1900s, Annie Jump Cannon and the Harvard Observatory Classification team produced the Henry Draper Catalogue (Cannon & Pickering 1918) of stellar classifications. They separated spectra into a series of groups via the strength of features in the stellar spectrum. In 1920, Meghnad Saha developed the ionization equations (Saha 1921), which describe the relationship between the ionization states of atoms and temperature. Cecilia Payne subsequently analyzed the observational data and classifications using the Boltzmann distribution and Saha ionization equation, with Fowler and Milne's work providing the theoretical framework for applying these principles to stellar spectra (Milne 1921, Fowler & Milne 1923). Payne demonstrated that variations in spectral features correspond to differences in temperature. This revealed that hydrogen and helium are the most abundant elements in stars (Payne-Gaposchkin 1925) and that the classification leveraged the changing line strengths of hydrogen and helium with temperature. This classification has been historically important because it organizes the spectra in a way that traces a physical process. **Figure 2** shows a series of GALAH [GALactic Archaeology with HERMES (High-Efficiency and Resolution Multi-Element Spectrograph) spectra for stars with similar metallicity but different temperatures. The center line is one of the Balmer series lines, the hydrogen-alpha (Hα) line. This temperature range (shown in **Figure 2**) only represents a fraction of the range of the initial classification but showcases the impact of temperature on the hydrogen line. Cooler stars have weaker Hα lines, and the hotter stars show increasing line strength within this range.

In the 1950s, it became commonplace to observe standard stars, particularly bright reference objects observable from both hemispheres, to transform between different photometric standards. This enabled cross-calibration of the different filter systems used to observe stars to get consistent results in the analyses. This was achieved via a simple approach of fitting a line between bands, either first or second order, to provide coefficients and enable stars to be calibrated between colors [for reviews, see Bessell & Brett 1988 and Bessell 2005]. Calibrations on contemporary data that are achieved with data-driven frameworks serve a similar purpose of homogenization, albeit over more variables and for more stars.

It has been recognized since the early visual classification that automation is necessary to handle large datasets and for studies of the Milky Way's structure and evolutionary history (e.g.,

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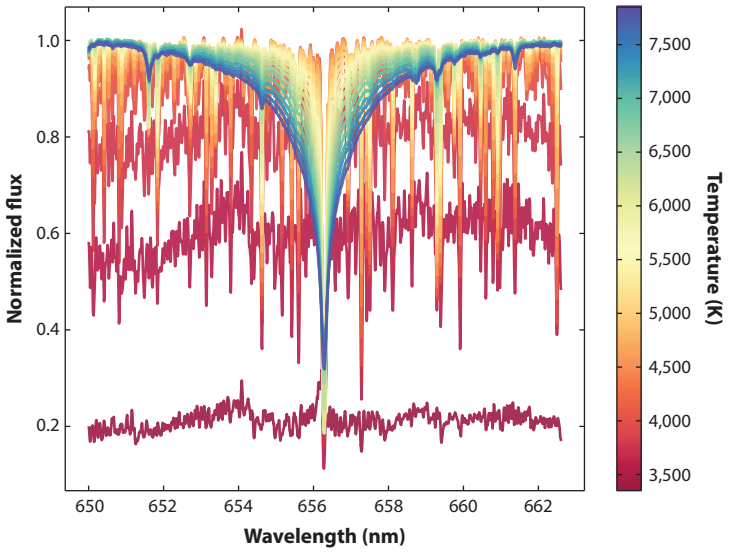


Figure 2

A sample of GALAH red giant spectra across $3{,}500 < T_{\rm eff} < 8{,}000$ K, for stars with $\log g < 4$ and [Fe/H] ~ 0. The region is centered on the hydrogen-α absorption line. This shows the impact of the $T_{\rm eff}$ on line strength; the hottest star within this range has the most prominent hydrogen-α line. Abbreviation: GALAH, GALactic Archaeology with HERMES (High-Efficiency and High-Resolution Multi-Element Spectrograph).

Schmidt-Kaler 1979, LaSala 1988). Early work by Malyuto & Shvelidze (1994) implemented a stepwise linear regression to learn classification, sorting reference stars into groups based on their stellar parameters ($T_{\rm eff}$, $\log g$, and [Fe/H]) using a modest set of high-resolution reference objects. This work aimed to determine a physical parameterization of stars with comparable accuracy to that achieved by using ab initio physical models and circumvented the need for labor-intensive model atmosphere analysis. Efforts undertaken in the 1990s in the pursuit of automating spectral classification proceeded alongside the growth of data volumes from multiobject fiber positioners, which had begun to be applied in observations (Lewis et al. 2002), and the recognition that data volumes would increase rapidly.

Pioneering directions set by von Hippel et al. (1994) were built upon and, as early as the late 1990s, artificial neural networks were being implemented to map stellar spectra to classifications, and principal component analysis (PCA) was deployed to classify spectra Bailer-Jones et al. (1997, 1998). Similarly, Snider et al. (1999) adopted neutral networks to estimate stellar effective temperature ($T_{\rm eff}$), surface gravity ($\log g$), and metallicity ([Fe/H]). They found these approaches to be successful and seamless for deriving these three labels at $R = 2{,}000$ as well as low SNR spectra, and for outlier identification.

The numbers of stellar spectra have grown substantially from the beginning of this century, driven by observational and instrumental capability, as well as computational support for efficient processing. The twenty-first century is the era of large surveys and large collaborations, with public data and public spectra for hundreds of millions to billions of objects. The catalogs of measurements from these surveys represent international and multi-institutional efforts that run for decades.

Alongside this observational data growth, the theory of stellar atmospheres and stars has steadily progressed. However, the theory remains largely limited in the domain of large-scale analyses by simplifying assumptions. These include one-dimensional, plane-parallel atmospheres

Neural networks: a computational modeling framework with layers of connected nodes used to encode complex relationships between input and output variables

Principal component analysis (PCA): a linear transformation that creates a subset of uncorrelated variables that explain the majority of variance in the original dataset

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that assume solar-scaled abundance ratios with mixing length factor approximations (e.g., Mihalas 1978, Asplund 2005). Developments in more comprehensive modeling are driving larger-scale three-dimensional non-local thermodynamical equilibrium (NLTE) efforts with a view toward future survey data (see the **Supplemental Text**).

Given the empirical nature of astronomy, data-driven models represent a natural direction for analysis. They offer flexibility and large-scale inference where traditional or theoretical models alone remain constrained. The continued growth of data-driven approaches will shape the future of the field, particularly as we enter an era of even larger missions, including space-based ones, that connect scales from exoplanets to galaxies. As data-driven methods mature, different subfields of astronomy will become increasingly interconnected.

3. THE LANDSCAPE OF SPECTROSCOPIC SURVEYS

More than 15 million stellar spectra are currently available from a multitude of ground-based surveys (R = 1,800–115,000). In addition, more than 200 million low-resolution (R = 20–100) and medium-resolution (R = 11,500) spectra have been observed by *Gaia* (e.g., De Angeli et al. 2023, Gaia Collab. et al. 2023, Katz et al. 2023).

Driving the rapid uptake of stellar spectra has been the development and propagation of multiobject spectrographs, the first of which began operation on the 1.2-m UK Schmidt Telescope at Siding Spring Observatory (Watson et al. 1992, Steinmetz et al. 2006), and expanded significantly with the Two Degree Field (2dF) instrument on the Anglo-Australian Telescope (AAT). The 2dF and robotic fiber positioner enabled 400 simultaneous observations in a two-degree field, and though developed for galaxy mapping (Lewis et al. 2002) has delivered millions of stellar spectra in ensembles of programs (e.g., Colless et al. 2001, Preibisch et al. 2002, Croom et al. 2004, Hilditch et al. 2005, Freeman 2010, Ness et al. 2013, Dobbie et al. 2014, De Silva et al. 2015, Lidman et al. 2016, Martell et al. 2017, Farrell et al. 2019, Li et al. 2019, Huang et al. 2025, Lowe et al. 2025). Multiobject spectrographs with automated fiber positioners are now commonplace and used in several of the surveys in **Table 1**. Note for the databases the total spectra are not unique stars and may include nonstellar spectra (e.g., galaxy, quasar).

The wavelength and spatial coverage, resolution, and scientific goals of different surveys are diverse, but they have common anchors. The Apache Point Observatory Galactic Evolution Experiment (APOGEE; R = 22,500) is a mapping survey done at infrared wavelengths (∼1.5−1.7 μm) designed to characterize the abundances and orbital properties, in concert with *Gaia*, of 0.7 million stars, in the $T_{\rm eff}$ range of 3,000 to 7,000 K (and primarily red giant stars) across the disk and bulge (Majewski et al. 2017). APOGEE spectra enable measurement of 18 elements (C, N, O, Na, Mg, Al, Si, S, K, Ca, Ti, V, Cr, Mn, Fe, Co, Ni, and Ce) with precisions of 5%, on average. The GALAH survey is a large-scale spectroscopic program at optical wavelengths (∼472−490, 564−587, 648−673, 759−789 nm) using the HERMES instrument on the AAT. GALAH has provided data to test chemical tagging—the premise of reconstructing stellar birth sites using their vector of abundances—via observations of almost 1 million stars with optical spectra, across the $T_{\rm eff}$ range of 3,000 to 8,000 K (and primary main sequence stars), delivering up to 28 elements (Li, C, O, Na, Mg, Al, Si, K, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Rb, Y, Mo, Ru, Ba, La, Nd, Sm, and Eu) at a precision of 15% on average (De Silva et al. 2015, Buder et al. 2025). The Gaia–European Southern Observatory Survey (Gaia-ESO; R = 20,000, 47,000) observing across optical wavelengths (400–682, 848–898 nm) provides high-resolution spectroscopic data for over 100,000 stars, also observed by *Gaia*, to study the formation of the Milky Way Galaxy's disk, bulge, and halo, reporting elements such as Li, C, N, O, Na, Mg, Al, Si, S, Ca, Sc, Ti, V, Mn, Fe, Co, Ni, Y, Zr, Ba, La, Ce, Nd, and Eu (Gilmore et al. 2012, 2022). This survey

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Table 1 Summary of some of the large spectroscopic surveys and instruments with data archives and their properties

Survey	Resolution (R)	Number of stars	Epoch of observations/ data release	Reference(s)
S^5	~1,300, 11,000	~100,000	2018–present	Li et al. 2019
PIGS	~1,300, 11,000	~13,000	2017–2024	Arentsen et al. 2020
ARGOS; A2A	~11,000	~28,000	2009–2013; 2021	Ness et al. 2013; Wylie et al. 2021
BRAVA	~4,200	~10,000	2005–2008 (DR:2012)	Kunder et al. 2012
H3	~32,000	~325,000	2020–present	Conroy et al. 2019
Gaia-ESO	~20,000, 47,000	>110,000	2012–2022	Gilmore et al. 2012
SEGUE	~1,800	~240,000	2004–2009 (DR7)	Yanny et al. 2009
RAVE	~7,500	~450,000	2003–2017 (DR5)	Steinmetz et al. 2006
GALAH	~28,000	~900,000	2013–2025 (DR4)	Buder et al. 2025
APOGEE	~22,500	~700,000	2011–2021 (DR17)	Majewski et al. 2017
LAMOST	~1,800	$>10 \times 10^6$	2011–present (DR8:2025)	Zhao et al. 2012
DESI	~5,000	~400,000 ($>5 \times 10^6$ planned)	2019–present (EDR:2024)	Dey et al. 2019, DESI Collab. et al. 2024
Gaia RVS DR3	~11,500	$\sim 1 \times 10^6$	2014–2022 (DR3)	Recio-Blanco et al. 2023
Gaia XP DR3	~20–100	$\sim 220 \times 10^6$	2014–2022 (DR3)	De Angeli et al. 2023
Gaia RVS DR4	~11,500	$\sim 250 \times 10^6$	2026 (DR5:2030)	TBD
Gaia XP DR4	~20–100	$\sim 2 \times 10^9$	2026 (DR5:2030)	TBD
SDSS V MWM	~22,500, 1,800	7×10^6 planned	2025–present (DR19:2025)	Kollmeier et al. 2025
4MOST	~5,000, 20,000	$>7 \times 10^6$ planned	Future	de Jong et al. 2016
WEAVE	~5,000, 20,000	$5-10 \times 10^6$ planned	Future	Bonifacio et al. 2016
MOONS	5,000, 9,000, 19,700	>1 million (planned)	Future	Bonifacio et al. 2016
ESO Survey archives	**Resolution**	**Number of spectra**	**Epoch**	**Reference(s)**
XSHOOTER	~4,000–17,000	>295,000 (>12,500 unique)	2009–present	Vernet et al. 2011, Verro et al. 2022
UVES	~40,000–110,000	>240,000 (>16,000 unique)	2000–present	Mayor et al. 2003
GIRAFFE	~5,500–65,000	>1,300,000 (>385,000 unique)	2003–present	Pasquini et al. 2002
HARPS	~110,000	>350,000 (>10,000 unique)	2003–present	Mayor et al. 2003
ESPRESSO	~110,000	>50,000 (>1,500 unique)	2018–present	Pasquini et al. 2009
FEROS	~48,000	>100,000 (>27,000 unique)	1998–present	Kaufer et al. 1999

Abbreviation: TBD, to be determined.

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emphasizes observing sets of calibration or reference objects (see Section 4). The Radial Velocity Experiment (RAVE; $R = 7{,}500$) obtained spectra in the calcium triplet region (∼841−880 nm) to measure radial velocities RVs and a small number of element abundances for ∼450,000 stars. This enabled a kinematic mapping of the Milky Way's disk and halo populations, allowing abundance measurements of 7 elements (O, Mg, Al, Si, Ca, Fe, Ni), as well as identification of stars with chromospheric emission via the core of the calcium triplet (Steinmetz et al. 2006, Casey et al. 2017). The Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST; $R = 1{,}800$) survey obtains spectra for millions of stars with a broad wavelength coverage (∼370−900 nm) to study the Milky Way's overall chemodynamics and structure, with up to 16 elements now extractable from these data (Zhao et al. 2012).

Upcoming spectroscopic surveys include those by the Multi-Object Optical and Near-infrared Spectrograph (MOONS; $R = 5{,}000$, 20,000) (Cirasuolo et al. 2014), the 4-meter Multi-Object Spectroscopic Telescope (4MOST; $R = 5{,}000$, 20,000) (de Jong et al. 2016), and the William Herschel Telescope Enhanced Area Velocity Explorer (WEAVE; $R = 5{,}000$, 20,000) (Bonifacio et al. 2016). Additional surveys targeting metal-poor populations include the Southern Stellar Stream Spectroscopic Survey (S^5) (Li et al. 2019), the Hectochelle in the Halo at High Resolution Survey (H3) (Conroy et al. 2019), and the Pristine Inner Galaxy Survey (PIGS) (Starkenburg et al. 2017, Ardern-Arentsen et al. 2024, Sestito et al. 2024). The Sloan Digital Sky Survey V's (SDSS V) Milky Way Mapper (MWM) survey builds on previous generations of SDSS. This is part of a mapping effort with the APOGEE ($R = 22{,}500$) and Baryon Oscillation Spectroscopic Survey (BOSS; $R = 1{,}800$) (Dawson et al. 2013) spectrograph to obtain spectra on millions of stars across all stellar types and across the plane of the disk to continuously and contiguously trace the components of the Galaxy (Kollmeier et al. 2025). MWP's flagship survey, Galactic Genesis, was borne out, in part, by the higher precision achieved by data-driven labeling. The *Gaia* mission maps hundreds of millions of stars across the Milky Way Galaxy at low- and medium-resolution ($R = 20$–100 and $R = 11{,}500$) (Gaia Collab. et al. 2016). In addition to these large surveys, telescope databases house millions of stellar spectra from $R = 1{,}800$–$115{,}000$ that can be accessed by the community. Some facilities and archives include the European Space Agency, ESO, Keck Observatory, Gemini observatory, AAT, Very Large Telescope, SDSS, Hobby–Eberly Telescope, and the NOIRLab (National Optical-Infrared Astronomy Research Laboratory) Astro Data Archive. Projects such as AMBRE (Archéologie avec Matisse: aBondances dans les aRchives de l'ESO), using HARPS (High Accuracy Radial velocity Planet Searcher) data, take archival spectra and determine consistent stellar properties for public catalog releases (De Pascale et al. 2014).

The different scientific goals of survey programs connect to the unique strengths and capability of each observatory and instrumentation. The data from these ensemble surveys have revealed a vast amount of information about the Milky Way, stellar populations, ecosystems, and stellar physics. Additionally, these surveys are revolutionizing our understanding of the nature of and information content in stellar spectra itself.

Large surveys have facilitated the development of data-driven approaches through the availability of both spectra and stellar labels, which are delivered by diverse and international teams. The open science efforts of collaborations like Astropy have similarly supported community progress (Astropy Collab. et al. 2013). The nature of international collaborative efforts, public data, and open-source software has had a huge impact and has been immensely beneficial to astronomy. The efforts of the teams that deliver measurements from spectra such as stellar parameters and abundances have advanced many components of the way analysis is done, as well as the completeness of theoretical models (see Recio-Blanco et al. 2006, 2023; Shetrone et al. 2015; García Pérez et al. 2016; Holtzman et al. 2018; Heiter et al. 2021; Smith et al. 2021; Buder et al. 2025; C. Li et al. 2024). So-called value added catalogs are often included in data releases, which invite and

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encourage the community to develop and deploy their own tools on the available data abundances. Large surveys now also include stellar parameter and abundance measurements delivered by data-driven approaches in their official catalogs for part or all of the survey (e.g., Buder et al. 2018).

Asteroseismology: the study of oscillation of stars, which encodes information about stellar internal structure and properties including radius, mass, and age

4. INGREDIENTS OF A DATA-DRIVEN MODEL

Data-driven frameworks built for differential analysis, classification, spectral decomposition, and generation of stellar spectra can be performed in an unsupervised way, without requiring descriptive labels. Therefore, models can be built with only stellar spectra and/or photometry, alongside computational capacity. However, labels are needed to interpret the variations encoded in spectra. The two main ingredients of a data-driven model built for the purpose of label transfer are therefore reference objects with high-fidelity labels and high-fidelity observational data—usually the spectra.

The labels typically derived from spectra include surface temperature $T_{\rm eff}$, surface gravity $\log g$, metallicity [Fe/H], and individual abundances [X/Fe], as well as micro- and macroturbulence. These are not direct observables like photometric color, apparent magnitude, and stellar flux. These labels serve as a means to transform the information encoded in photons from photospheres into a theoretical physical framework. The values for $T_{\rm eff}$ and $\log g$ express the state of a stellar spectrum of evolving stars; they are parameters that set the characteristics of the layers in the model atmosphere, and the [Fe/H] and [X/Fe] are the material with which a star was born, modulo any changes over time due to atomic diffusion, dredge-up, planet engulfment, or binary interaction (e.g., Masseron & Gilmore 2015, Dotter et al. 2017, Liu et al. 2019, Oh et al. 2018, Souto et al. 2018).

Reference objects are key. They set the basis for a ground-truth stellar parameter and abundance (label) scale. In the context of data-driven models, they set the scale for all other spectra to be labeled by the model that is built from the reference objects. These are not only ingredients for data-driven models but may also serve as the calibration objects for ab initio physical models built from first principles. Reference objects can test the fidelity of stellar physics used to generate theoretical representations.

Reference objects for data-driven models can be as simple as high SNR data in any survey or be expensive targets that are studied extensively by multiple missions with independent information. The Gaia-ESO has a set of stars specifically observed as a reference set, the *Gaia* benchmark stars (e.g., Heiter et al. 2015, Hawkins et al. 2016, Jofré et al. 2017, Soubiran et al. 2024). These stars are observed at high resolution and SNR and have independent measurements of parameters, including interferometric angular diameters. Most surveys also target open and globular clusters to serve as conatal populations, as well as stars observed by asteroseismology with precision masses and ages (Feltzing 2016; Pinsonneault et al. 2018, 2025; Wittenmyer et al. 2018). Classes of objects such as red giants in eclipsing binaries are also a benchmark for asteroseismic samples (Aerts 2021, Beck et al. 2024).

Reference objects such as clusters are also extremely important for characterizing the trends in abundances across evolutionary state, as shown in work by Kos et al. (2025) with GALAH clusters (as shown in **Figure 3**). These trends may in part be astrophysical due to diffusion and dredge-up as well as due to inaccuracies in stellar models (Jofré et al. 2019). They can be calibrated out using novel data-driven modeling of the stellar labels (see Section 5.3). A list of example reference objects is provided in **Supplemental Table 1** in the **Supplemental Text**.

Typically, million-star surveys in different hemispheres have tens of thousands of stars in common (e.g., Wheeler et al. 2020, Buder et al. 2025, Zhang et al. 2024, Das et al. 2025). These stars in common provide the means of placing surveys directly on the same stellar parameter and

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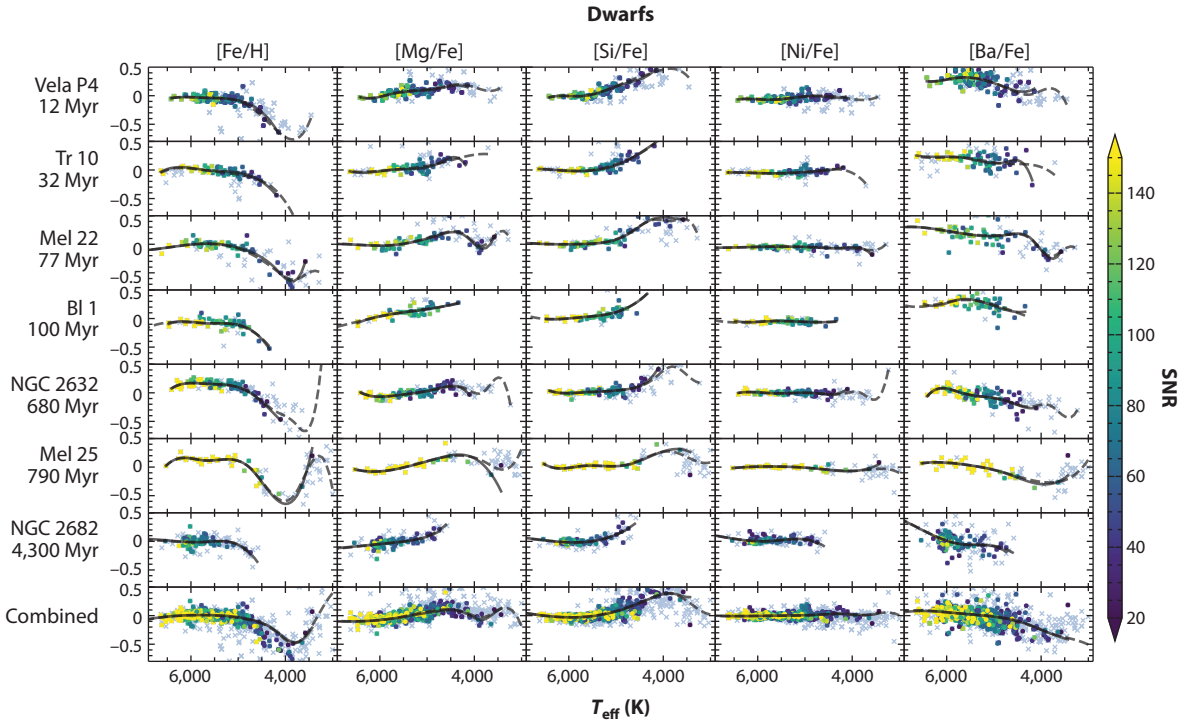


Figure 3

Element abundance trends as a function of effective temperature (T_{eff}) for stars in several clusters observed by GALAH, shown for a subset of elements. Element abundance–T_{eff} trends are present in the data, which are likely nonphysical and instead reflect systematic effects in the analysis. Figure adapted with permission from Kos et al. (2025) (CC BY 4.0). Abbreviation: GALAH, GALactic Archaeology with HERMES (High-Efficiency and High-Resolution Multi-Element Spectrograph).

abundance (label) scale with data-driven modeling approaches. These shared stars can be used as the reference objects and, therefore, to set a single label scale that is propagated to each large survey using models based on each survey's spectra. This approach propagates the same label scale among surveys, solving the problem of different label scales across individual surveys.

5. MODERN DATA-DRIVEN METHODS AND THEIR UTILITY

Data-driven methods in spectroscopy have grown in diversity and number over the past two decades. The advantages gained from modeling the data align with the requirements in the inquiry of nonstellar objects. This includes studies of the interstellar medium and dust, as well as signals of exoplanets and their atmospheres. Data-driven astronomy therefore very naturally connects domains both within astronomy and outside of it, interfacing with developments in machine learning and artificial intelligence.

5.1. Component Separation and Denoising Spectra

Data-driven processing can deliver betterized stellar spectra. Cleaned spectra that do not have residing nuisance signals are important in multiple applications. This includes exoplanet discovery,[4]

[4]For a review of data-driven methods in the exoplanet domain, we refer to the review by Burt et al. (2026, in this volume).

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exoplanet atmosphere analysis, and chemodynamical analysis, for example, learning the origin of the elements. All these pursuits require precision measurements.

Data-driven models can remove unwanted signals in spectra, enabling more accurate and precise measurements. Data processing models for this purpose use both time-domain data of single stars or the population data of many stars. They pool data to remove nuisance signals and extract those of interest at higher fidelity. For example, the `Wobble` package (Bedell et al. 2019) uses time-domain observations to decompose stellar spectra into components attributed to stellar and telluric contributions. This enables precision RV observations for exoplanet science. `ExCALIBUR` (Zhao et al. 2021) delivers precision-calibrated spectra for exoplanet science disentangled from a composite of other information by extracting the stellar signal from its ensemble of multiple contributions by building a model that incorporates instrumental systematics. Using a population of spectra, `MADGICS` (Marginalized Analytic Data-space Gaussian Inference for Component Separation) (Saydjari et al. 2023) jointly infers the different components in stellar spectra, by organizing the features into a set of primary contributions. This process isolates sky signals and instrument systematics, enabling more precise abundances and velocity measurements from the betterized spectra (Saydjari et al. 2025). Models of stars built from the data also enable discovery using the residual amplitudes of the data model. Interstellar signals from diffuse interstellar bands have been recently discovered and mapped in this way (e.g., McKinnon et al. 2024). Calibration models built for these tasks do not require any physical theory unlike models that are deployed for the purpose of labeling stellar spectra, which do rely on theoretical models of stellar spectra.

5.2. Labeling Stellar Spectra

Stars can be labeled within and between surveys using reference objects. The reference set of stars serves as the label basis and ground truth. The reference set of objects should be representative of the test set[5] (often a large survey set) that the model is deployed on, insofar as label-space domain, resolution, and wavelength coverage—but not necessarily data quality (i.e., SNR, depending on the modeling framework and whether or not flux uncertainties are input to the model). Although these methods have been developed and applied in the spectroscopy domains largely across $T_{\rm eff} = 3{,}000$–$7{,}000$ K, the methods themselves are general and not restricted to this range. A model built from the data using stars common between surveys—whereby labels are inherited from one survey and spectra from another—enables surveys to be placed directly on the same label scale. This solves the problem of different surveys having different label scales; as long as one is adopted as ground-truth, this can be propagated to other wavelength regions and other stars.

Data-driven labeling or label transfer has been achieved with numerous different approaches and has been applied to many surveys. Method choices are motivated by different goals and built to serve different data. Typically, these approaches work with continuum-normalized (or continuum-rectified; the instrument profile is being removed), velocity-corrected stellar spectra, with uncertainties reported at each wavelength. Although not a requirement, continuum normalization and velocity correlation deliver the data in a format that models take as convenient input for label transfer.

Different implementations of label transfer vary in both fundamental and nuanced ways. Although various criteria can be used to categorize these methods, they exhibit overlapping properties. Additionally, models that share fundamental principles may diverge in practice due to

[5] In this review, a reference set of objects is defined as the objects on which the model is applied; this is often large sets of survey data in which new labels and/or model spectra are determined for each star. (A reference set can also describe a set of stars used to validate the model.)

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specific implementation choices within their broader frameworks. One significant way models differ is in their flexibility and interpretability. This distinction can be either inherent (e.g., using a neural network mapping or decision tree to learn a mapping between ensemble spectra and labels in a discriminative way or to separate objects into classes represented by labels) or a choice within a broader statistical framework (e.g., model selection in a probabilistic generative approach, to generate data from inferred labels).

Regularization: a penalty term incorporated into an optimization to penalize complex solutions and set constraints on the optimization and, subsequently, solution

A common differentiator of modeling approaches is also whether they are generative or discriminative, and some incorporate elements of both. Model complexity, or completeness in a statistical sense, manifests in whether or not they take into account label as well as flux uncertainty, and if and how they handle missing and noisy data. Some models integrate information from stellar physics, and some include formalisms for the purpose of stability and robustness (e.g., regularization). Models may also project data into latent or encoded spaces to emphasize the sources of variation linked to the labels or classes in classification, which can lead to more accurate or precise inference. Although most data-driven methods for spectroscopy leverage the ensemble of combined flux to build models, which can be useful to break degeneracies (e.g., if individual data values have multimodal correlations between the labels that require convolution of the data to learn a unique label mapping), simpler methods exist that treat the flux at each wavelength independently.

Supplemental Table 2 in the **Supplemental Text** summarizes some of the data-driven modeling approaches that are available in stellar spectroscopy, including both supervised models used for label transfer and unsupervised methods for spectral decomposition or querying the information encoded in spectra. Subsets of these models are detailed below to motivate why different choices might be made and how these are implemented.

5.2.1. Flux as a linear combination of coefficients and labels. The perhaps simplest data-driven label-transfer approach is `The Cannon` (Ness et al. 2015), named after Annie Jump Cannon (see Section 2.2). Its description sets out a general framework via which data-driven label transfer works. Like other supervised automated learning methods, it involves a training step, in which reference stars are used to train a model, and a test step, in which that model is applied to infer new labels from stellar flux. This procedure has similarities with earlier tools that use synthetic spectra rather than the data itself to train on (e.g., Bailer-Jones et al. 1998, Recio-Blanco et al. 2006, Re Fiorentin et al. 2007, Koleva et al. 2009), as well as some methods that have adopted empirical libraries of spectra (see Soubiran et al. 1998, Prugniel et al. 2011, Wu et al. 2011). One important aspect of working in the space of the data that differentiates data-driven models from those that use synthetic spectra pertains to the model flexibility. The model only has to be as flexible as required by the physical constraints of where stars are located in the parameter or label space. This enables a relatively simple model to be used in `The Cannon`.

The model that has typically been used in `The Cannon` is a simple and interpretable second-order polynomial that relates labels to the flux via an algebraic equation of coefficients multiplied by the first-order cross and square terms of the labels. In first-order linear terms, the model is therefore directly interpretable; the coefficients represent the gradient of the flux with the label at hand, offering physical insight into the spectral sensitivity. This simplicity also makes `The Cannon` very fast. In short, in the training step, it uses a weighted least-squares optimization (see Hogg et al. 2010 for a pedagogical description of model fitting) to learn the relationship between flux and labels from a reference set of stars. It does this by building an independent polynomial model of the flux at each wavelength as a function of the labels. At test time, labels for new spectra are inferred by maximizing the likelihood under the trained flux model. The same trained model can also be used to generate synthetic spectra by evaluating the polynomial flux model at specified label values (see **Figure 4**). The second-order polynomial combination of coefficients and labels

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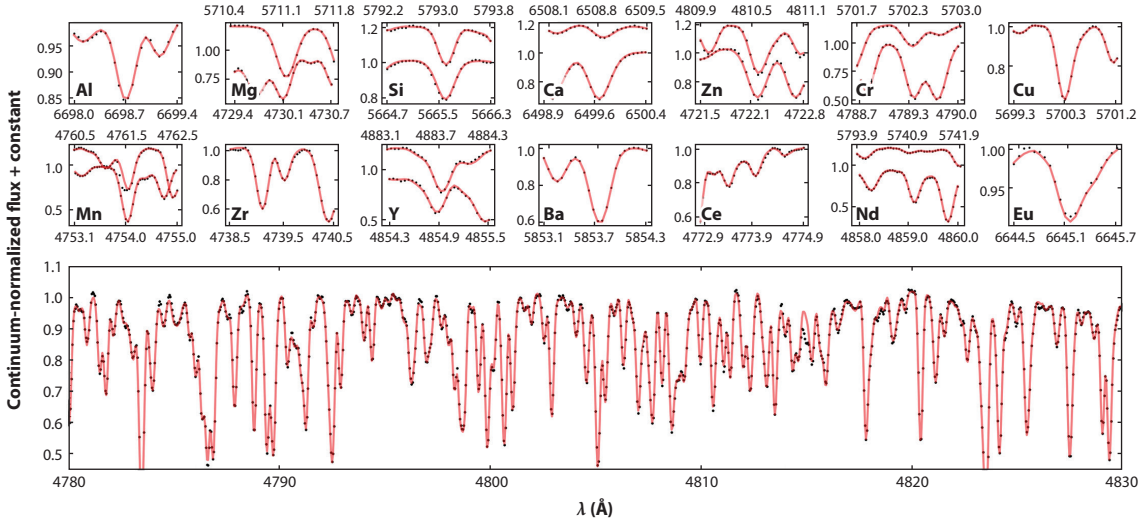


Figure 4

The GALAH spectral region showing, at bottom, a narrow wavelength region and, at top, the regions corresponding to different elements being inferred using `The Cannon`. The black points show the data, and the red line is the model generated by multiplying `The Cannon`'s model by the inferred labels. This test of theoretical and observed spectra is an additional test to confirm that the labels faithfully generate the spectra, including in the individual element absorption features. Figure adapted with permission from Manea et al. (2024) (CC BY 4.0). Abbreviation: GALAH, GALactic Archaeology with HERMES (High-Efficiency and High-Resolution Multi-Element Spectrograph).

that has been successful for many applications represents a choice in implementation and can be replaced within the same framework, e.g., by a neural network (see El-Badry et al. 2018).

The general framework of `The Cannon` assumes that there are $N_{\rm ref}$ reference objects, n, with continuum-normalized flux that has been doppler corrected to the rest frame, $f_{n\lambda}$ at wavelength λ. Each of the n training spectra has K labels ℓ_{nk}, expressed in the model in a label vector $\boldsymbol{\ell}_n$. For any star n at any wavelength λ, the flux $f_{n\lambda}$ can be described as a function of the labels ℓ_{nk}. The labels are numbers that describe the way flux varies, which can include stellar parameters and abundances, turbulent and rotational velocities, and age and asteroseismic parameters (e.g., $T_{\rm eff}$, $\log g$, [Fe/H], [X/Fe], $v_{\rm mic}$, age, $v \sin i$). This can be written as follows, whereby a coefficient vector $\boldsymbol{\theta}_\lambda$ allows the prediction of the flux at every pixel $f_{n\lambda}$ for a given label vector $\boldsymbol{\ell}_n$:

$$f_{n\lambda} = g(\boldsymbol{\ell}_n|\boldsymbol{\theta}_\lambda) + \text{noise}. \quad 1.$$

Using this formalism, `The Cannon` relates stellar flux to its descriptive labels. Specifically, in `The Cannon`, the following functional form is adopted in which the model is a linear function of the label vector $\boldsymbol{\ell}_n$, where $\boldsymbol{\theta}_\lambda$ are the model coefficients of the spectra at each λ:

$$f_{n\lambda} = \boldsymbol{\theta}_\lambda^T \cdot \boldsymbol{\ell}_n + \text{noise}. \quad 2.$$

The noise term in the model is described by a quadrature sum of the measurement uncertainty variance $\sigma_{n\lambda}^2$ and the intrinsic model variance at each wavelength, s_λ^2, which captures additional sources of scatter in the data. This model assumes that the noise model is described as noise $= [s_\lambda^2 + \sigma_{n\lambda}^2]\,\xi_{n\lambda}$, where each $\xi_{n\lambda}$ is a Gaussian random number with zero mean and unit variance. This scatter of the model is solved for at training time as part of the optimization. This term is

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important in setting the weighting of the wavelength values, which, though trained independently and thereby result in a model at every wavelength, are used in concert at test time to infer labels. The single-wavelength log-likelihood function becomes

$$\ln p(f_{n\lambda} \mid \boldsymbol{\theta}_\lambda^T, \boldsymbol{\ell}_n, s_\lambda^2) = -\frac{1}{2} \frac{[f_{n\lambda} - \boldsymbol{\theta}_\lambda^T \cdot \boldsymbol{\ell}_n]^2}{s_\lambda^2 + \sigma_{n\lambda}^2} - \frac{1}{2} \ln(s_\lambda^2 + \sigma_{n\lambda}^2). \quad 3.$$

The vector $\mathbf{f}_\lambda$ is the set of spectral flux values for all N objects at the one wavelength λ. The coefficients $[\boldsymbol{\theta}_\lambda, s_\lambda^2]$ are determined by optimizing the likelihood in Equation 3 over all reference objects. In general, it would also be possible to include priors or apply probabilistic inference. The optimization of the model parameters $\theta_{\lambda k}$, at fixed scatter s_λ^2 is a (weighted least squares) linear-algebra operation. If the model choice were restricted to a first-order linear form, then the resulting optimization to infer labels is also convex. However, the polynomial form is nonlinear in the labels and, therefore, does not have a closed form solution for inferring the labels. At test time the labels for each star are inferred through a simultaneous optimization over all label parameters, with the trained coefficients of the model $\boldsymbol{\theta}_\lambda$ and intrinsic scatter s_λ^2 held fixed.

In the example provided by Ness et al. (2015), it is assumed that for any star, n, at any pixel, λ, the flux $f_{n\lambda}$ can be described as some smooth function of the star's labels ℓ_{nk}. In the training step of `The Cannon`, both the fluxes for the reference objects, $f_{n\lambda}$, and the corresponding stellar labels, $\boldsymbol{\ell}_n$, are known. Subsequently, the coefficients and scatter of the spectral model can be solved for

$$\boldsymbol{\theta}_\lambda, s_\lambda \leftarrow \underset{\boldsymbol{\theta}_\lambda, s_\lambda}{\operatorname{argmax}} \sum_{n=1}^{N} \ln p(f_{n\lambda} \mid \boldsymbol{\theta}_\lambda^T, \boldsymbol{\ell}_n, s_\lambda^2). \quad 4.$$

The vector $\boldsymbol{\ell}_n$, which is quadratic in the labels, is described as

$$\begin{aligned} \boldsymbol{\ell}_n \equiv & [1, \ell_{n1} - \bar{\ell_1}, \ell_{n2} - \bar{\ell_2}, \ldots, \ell_{nK} - \bar{\ell_K}, \\ & (\ell_{n1} - \bar{\ell_1})(\ell_{n1} - \bar{\ell_1}), (\ell_{n1} - \bar{\ell_1})(\ell_{n2} - \bar{\ell_2}), \ldots, \\ & (\ell_{nK} - \bar{\ell_K})(\ell_{nK} - \bar{\ell_K})]. \end{aligned} \quad 5.$$

The flux can be described as a polynomial function of the stellar labels ℓ_{nk}, where $k = 1, \ldots, K$ indexes the labels for star n. Expanding this dependence explicitly, the flux at wavelength λ can be written as

$$\begin{aligned} f_{n\lambda} = & \theta_\lambda^0 && \text{(constant term)} \\ & + \theta_\lambda^k \ell_{nk} + \ldots && \text{(linear terms)} \\ & + \theta_\lambda^{kk} \ell_{nk}^2 + \ldots && \text{(squared terms)} \\ & + \theta_\lambda^{kk'} \ell_{nk} \ell_{nk'} + \ldots && \text{(cross terms)} \\ & + \text{noise}. \end{aligned}$$

In the test step of `The Cannon`, the spectral model and coefficients and scatter are used, $(\boldsymbol{\theta}_\lambda, s_\lambda^2)$, as per the training step, but taking the entire spectrum of survey star m, $f_{m\lambda}$, and optimizing for the

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labels of that star, we get

$$\{\ell_{mk}\} \leftarrow \underset{\{\ell_{mk}\}}{\text{argmax}} \sum_{\lambda=1}^{N_\lambda} \ln p(f_{m\lambda} \mid \boldsymbol{\theta}_\lambda^T, \boldsymbol{\ell}_m, s_\lambda^2). \quad 6.$$

To maximize interpretability and mitigate any subtle bias (though this is still minimized under the relatively inflexible model), for some use cases it is favorable to implement `The Cannon` using a first-order linear model instead of the quadratic function described. This is a good solution in restricted parameter spaces, or it can be used in a system of local (first-order) linear models across broader parameter spaces, albeit at more computational expense than a global model.

`The Cannon` models the flux as a function of the labels and coefficients at every wavelength independently, building a probability distribution of the flux at every wavelength. This is a distinguishing and simplifying aspect compared to many other modeling frameworks. It means that at test time, different parts of the model can be trivially excluded on a per-label basis if desirable. A typical motivation for doing this censoring is in inferring element abundances using only narrow windows across wavelengths centered at their absorption features. In practice, it is possible to simply set the coefficients to zero at the wavelengths associated with the desired labels outside of the windowed regions. However, in some cases, one might be well-advised to use the entire spectrum even for abundances, including oxygen (Ting et al. 2018a), or in blended regions in which abundance information may be distributed over many wavelength regions. This is particularly relevant in the case of cool stars, like M dwarfs, which are densely blanketed with atomic and molecular features that are blended even at high resolution. In addition, due to complexity in interaction with the overall atmosphere, elements may impact the entire atmospheric stellar structure and information about them expressed outside their transition wavelengths (e.g., Battaglia et al. 2008, Gustafsson et al. 2008, Starkenburg et al. 2010, Mucciarelli et al. 2012, Veyette et al. 2016).

In `The Cannon 2`, an extended version of `The Cannon`, Casey et al. (2016) incorporated L1 regularization into the training process. This encourages coefficient values to become zero. This reduces the risk of overfitting and means the model relies only on the most informative label coefficients at each wavelength.

`The Cannon` and related data-driven models appear to infer labels at higher precision than other methods (see, e.g., Casey et al. 2016, Ness 2018, Leung & Bovy 2019, Manea et al. 2024), even when restricted to the same window regions per element. This improved precision is presumably related to using the full error-weighted spectrum to infer each label and that the data-driven model is (by definition) a better generative match to the spectra than the ab initio models. The precision gain translates into the capacity to observe more stars to achieve a given precision goal—because the required observing time scales as $1/n^2$, where n is the factor by which the lower SNR spectrum achieves the same precision as a higher-SNR target. This motivates obtaining more stars at lower SNR such as is done in the Galactic Genesis program in SDSS V (Kollmeier et al. 2025), which is mapping almost 3 million stars continuously and contiguously across the dust-extincted disk at a relaxed requirement compared to previous generation surveys, of SNR > 35.

`The Cannon` has been deployed on numerous surveys with resolution from $R = 1{,}800$ to $R = 80{,}000$, including LAMOST (Ho et al. 2017a,b; Wheeler et al. 2020), APOGEE (Ness et al. 2016, 2018a; Hawkins et al. 2018; Behmard et al. 2019, 2025; Birky et al. 2020; Hasselquist et al. 2020; Sit & Ness 2020; Yuxi et al. 2022), CTIO (Cerro Tololo Inter-American Observatory)/Hydra spectra (Ray et al. 2022), GALAH (Buder et al. 2021, Nandakumar et al. 2022, Manea et al. 2024, Walsen et al. 2024), ARGOS (Abundances and Radial velocity Galactic Origins Survey; Wylie et al. 2021), Keck's High Resolution Echelle Spectrometer (HIRES; Behmard et al. 2019,

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Latent variable: a variable that represents the underlying structure or patterns in the data but is not a physical representation of it

Rice & Brewer 2020, Gussman & Rice 2024), RAVE with `The Cannon 2` (Casey et al. 2016), *Gaia* (e.g., Angelo et al. 2024, Rampalli et al. 2024, Das et al. 2025), Multi Unit Spectroscopic Explorer (MUSE; Wang et al. 2022, Ness et al. 2025), the Wide Field Spectrograph (WiFeS; Rains et al. 2024), XSHOOTER (Banks et al. 2024), Veloce (Banks et al. 2024), and *Kepler* (Ness et al. 2018b) (also see a related method in Sayeed et al. 2021). It also propagates very well down to low SNRs. When spurious features are present but do not correlate with labels, it can be robust to these. It has been shown to perform well, including for inference of 15 element abundances, from iodine-imprinted spectra from Keck HIRES (Gussman & Rice 2024). The generative nature of the model is a useful tool for providing model spectra for precision RV shifts (e.g., Nidever 2021). Other simple empirical methods are calibrations that relate the equivalent width of the calcium triplet to metallicity (e.g., Starkenburg et al. 2010, Vásquez et al. 2015).

5.2.2. Flux as a probabilistic mapping of latent variables and labels. Although temporally a descendant, in terms of design, `Lux` (Horta et al. 2024a) is an antecedent to `The Cannon` and in the simplifying case can reduce to its form (but `The Cannon` cannot become `Lux`). `Lux` (Horta et al. 2025) is a generative data-driven label-transfer method that accounts for noisy input data (both labels and spectra). The `Lux` framework builds a full probabilistic model of continuum-normalized Doppler-shifted stellar flux and stellar labels using a linear latent variable model (where the linear aspect is a choice) to amplify signals that are relevant to label inference. Other uses of latent variable data-driven modeling are discussed by Nolan et al. (2006), Hall et al. (2019), and Eilers et al. (2022b).

`Lux` can generate spectra from inferred labels and can reconstruct spectra or infer labels from observed flux. Like `The Cannon`, the model takes as input the flux and labels, and the flux uncertainty. In addition however, it takes into account the uncertainty of the labels. A subsequent advantageous utility of `Lux` compared to other models is that it allows the construction of a comprehensive training set of objects with ensemble labels of heterogeneous fidelity (both in label and flux). It also adopts formalisms for stability and requires user selection of hyperparameters, including for regularization and number of latent variables.

In `Lux`, the labels $\boldsymbol{\ell}$ are considered to be generated by a function of a set of latent parameters $\mathbf{z}$, projected onto the physical label space by a matrix $\mathbf{A}$. Similarly, the flux $\mathbf{f}$ is generated by the latent parameter vector multiplied by a transformation matrix $\mathbf{B}$ to generate normalized flux. The dimensionality of the latent space is specified by parameter P. For the n reference objects, this becomes in both cases for each star n,

$$\boldsymbol{\ell}_n = \mathbf{A}\,\mathbf{z}_n + \text{noise}, \tag{7.}$$

and

$$\boldsymbol{f}_n = \mathbf{B}\,\mathbf{z}_n + \text{noise}. \tag{8.}$$

Here, the noise is assumed to be Gaussian, and the variances are taken as input. Under the assumption that the per-star (n) likelihood is conditionally independent, the full likelihood of the joint data for a set of N stars becomes the product of the per-star likelihoods:

$$\mathcal{L}(\mathbf{A}, \mathbf{B}, \{\mathbf{z}_n\}_N, s_\lambda) = p(\{\boldsymbol{\ell}_n\}_N, \{\boldsymbol{f}_n\}_N \mid \mathbf{A}, \mathbf{B}, \{\mathbf{z}_n\}_N, s_\lambda) = \prod_{n=1}^{N} p(\boldsymbol{\ell}_n, \boldsymbol{f}_n \mid \mathbf{A}, \mathbf{B}, \mathbf{z}_n, s_\lambda). \tag{9.}$$

Here, $\mathbf{s}$ is the vector representing the scatter term at each wavelength λ, which captures unaccounted for sources of variance in the relationship between the flux and the labels given the model.

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Given the likelihood function, similar to `The Cannon`, it is possible to proceed probabilistically or optimize the likelihood directly. In doing so, `Lux` infers stellar labels from stellar spectra given a reference set of stars to learn the model ($\mathbf{A}$, $\mathbf{B}$, $\mathbf{z}_n$, and $\mathbf{s}$) and a test set of stars to which the model is applied to infer the labels $\boldsymbol{\ell}$. The model is improved with the inclusion of a ridge regression regularization term (represented by the second term, where Ω serves as a regularization coefficient to control the contribution of this term), and the optimization becomes

$$g(\mathbf{A}, \mathbf{B}, \{\mathbf{z}_n\}_N, \boldsymbol{s}_\lambda) = \sum_n^N \left[\ln p(\boldsymbol{\ell}_n \mid \mathbf{A}, \mathbf{z}_n) + \ln p(\boldsymbol{f}_n \mid \mathbf{B}, \mathbf{z}_n, \boldsymbol{s}_\lambda) + \Omega \sum_p^P \mathbf{z}_{pn}^2 \right]. \quad 10.$$

`Lux` generates both flux and labels, and it is possible to use the matrices $\mathbf{A}$ and $\mathbf{B}$ to calculate the derivative of flux with respect to the labels and vice versa, thereby making the framework interpretable in the sense of understanding where the information pertaining to the labels is present in wavelength. For APOGEE red giant spectra ($4{,}000 < T_{\rm eff} < 5{,}500$ K, $0 < \log g < 3.5$) at SNR = 50, `Lux` achieves a precision of <0.01 dex (3%) in elements O, Mg, Si, Ca, and Mn and a precision of <0.02 dex (5%) in elements C, N, Al, Fe, and Ni.

In summary, `Lux` serves multiple purposes, including label inference, cross-survey label transfer, spectrum generation, and classification while incorporating uncertainties in both labels and flux. By including the label uncertainties in the model, `Lux` relaxes the strict requirements of methods like `The Cannon`, which assume that training labels are perfectly known; `The Cannon` model suffers from labels with high uncertainties. In addition, `Lux` handles the missing label and flux without the arbitrary workarounds required in other methods. Although `Lux` is more complex than `The Cannon`, it is also more comprehensive and statistically complete. The model is implemented using `JAX` (Bradbury et al. 2021), ensuring computational efficiency. The linear transformations it uses can be extended to more flexible nonlinear mappings. Because `Lux` accommodates ensemble label and spectral inputs from different sources using a shared latent-space embedding, it offers a path toward holistic modeling and more accurate label predictions across heterogeneous datasets.

5.2.3. Flux as a nonlinear function regularized by physical models. The data-driven version of `The Payne` (Xiang et al. 2019) is a flexible generative modeling approach based on `The Payne` (Ting et al. 2019), which is named after Cecilia Payne (see Section 2.2). This method takes as input information from physical stellar models to constrain the model learned from the data (also see Zhang et al. 2024). Its flexible model is a neural network that learns the mapping between continuum-normalized, velocity-shifted, observed spectra and the labels that describe it. Key to the method is the inclusion of constraints from synthetic spectra in the label inference to prohibit the full reliance on the data to inform where the labels manifest in the spectra, thereby avoiding learning from correlations. This enables both accurate and precise labels informed by constraints from ab initio models. The `Data-Driven Payne` thereby reduces limitations in data-driven frameworks pertaining to the use of correlated information to infer information rather than sources of causation by including gradient spectra from the stellar atmosphere models in the objective function optimization. Gradient spectra are the derivatives of ab initio spectra with respect to each stellar label across wavelength. These gradient spectra quantify how sensitively the flux responds to changes in individual stellar parameters or abundances and, therefore, set prior information about where information pertaining to each label resides in observed spectra, according to theoretical models. Subsequently, these inform where the spectral flux varies in response to each label.

The `Data-Driven Payne` takes as input the stellar labels, observed normalized flux, gradient spectra, and flux uncertainties. A weighting is applied for the regularization term using the

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Decision Tree: a machine-learning method with a series of connected branches that are used to make predictions using a series of rules or conditions

Gradient boosting: a method to improve model prediction accuracy by training an ensemble of models sequentially, whereby each new model minimizes the residuals (difference between the measured and predicted values) of the previous model

gradient spectra, which serves as prior information about where the spectral sensitivity to each label originates. The `Data-Driven Payne` constrains the model using physically motivated gradients, mitigating learning from noncausal correlations. For example, inferring abundance labels by leveraging the entire spectrum can lead to label predictions based on regions that vary with the label only indirectly, because of correlated information distributed across the spectrum.

The loss function (i.e., the objective function to minimize) used in the `Data-Driven Payne` is given by (see Zhang et al. 2024)

$$\mathcal{L}(\{\mathbf{f}_{\rm obs}(\lambda)\} \mid \mathbf{w}, \mathbf{b}) = \frac{1}{N_{\rm s}} \sum_{i=1}^{N_{\rm s}} \frac{(f(\lambda \mid \ell_{{\rm obs},i}) - f_{{\rm obs},i}(\lambda))^2}{\sigma^2_{{\rm obs},i}(\lambda)} + \sum_{j=1}^{N_{\rm r}} \mathbf{D}_{\rm scale} \cdot \sum_{k=1}^{N_{\rm l}} \left| f'(\lambda; \boldsymbol{\ell}_{\rm ref}) - f'_{\rm physical}(\lambda; \boldsymbol{\ell}_{\rm ref}) \right|, \quad 11.$$

where $f_{{\rm obs},i}(\lambda)$ is the observed normalized flux at wavelength λ for the ith training star, $f(\lambda|\ell_{{\rm obs},i})$ is the model flux at wavelength λ given the stellar label $\ell_{{\rm obs},i}$, $\sigma_{{\rm obs},i}(\lambda)$ is the observational uncertainty at wavelength λ for the ith star, $f'_{\rm physical}(\lambda; \boldsymbol{\ell}_{\rm ref})$ is the spectral gradient derived from stellar physical models for reference labels $\boldsymbol{\ell}_{\rm ref}$, $f'(\lambda; \boldsymbol{\ell}_{\rm ref})$ is the model-predicted spectral gradient at λ, $\mathbf{D}_{\rm scale}$ is a vector determining the weight of each label in the regularization, $N_{\rm s}$ is the number of training spectra, $N_{\rm r}$ is the number of reference stars, $N_{\rm l}$ is the number of labels, and $\mathbf{w}$ and $\mathbf{b}$ are the coefficient arrays to be optimized for a two-layer perceptron (2LP) neural network.

In the training process, the loss function is regularized by the gradient spectra from stellar atmospheric models, which appears in the second term of Equation 11. This restricts the flexible model from leveraging regions in the spectra corresponding to where labels impute variation.

A nonlinear neural network interpolation model of the form below maps the stellar labels that are inferred for each star to stellar spectra, following Ting et al. (2019) (where the index summation is written in Einstein notation):

$$\mathbf{f}_\lambda = \mathbf{w} \cdot \sigma(\tilde{\mathbf{w}}^i_\lambda \sigma(\mathbf{w}^k_{\lambda,i} \boldsymbol{\ell}_k + \mathbf{b}_{\lambda i}) + \tilde{\mathbf{b}}) + \bar{f}_\lambda, \quad 12.$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function, $\boldsymbol{\ell}$ are the labels, $\mathbf{w}$ and $\mathbf{b}$ are coefficient arrays to be optimized, i is the index for the number of neurons, k is the number of labels, and $\bar{f}_\lambda$ is a baseline function.

The `Data-Driven Payne` has been applied to numerous surveys. Xiang et al. (2019) report abundances for 16 elements in 6 million stars from LAMOST ($R = 1{,}800$), and Zhang et al. (2024) demonstrate the derivation of 12 elements from Dark Energy Spectroscopic Instrument (DESI) ($R = 5{,}000$) spectra. This has also been deployed using labels from APOGEE and a model from LAMOST to infer extragalactic abundances from resolved MUSE spectra (Wang et al. 2022) (see Section 5.2.6).

There are methods that extend the Payne to hotter temperature regimes (>7,000 K) with Zeta–Payne (Gebruers et al. 2022, Straumit et al. 2022). These rely on model grids for training and are not data driven, although they could be implemented similarly to the `Data-Driven Payne`.

5.2.4. Labels predicted from flux with boosted decision trees. The XGBoost decision tree method (Chen & Guestrin 2016) is a discriminative and computationally inexpensive approach that handles missing data and learns the nonlinear mapping between spectral features and stellar labels. It can estimate stellar parameters across a large parameter space. XGBoost is computationally efficient and has been used for data-driven label inference in very large surveys in particular. Decision trees enhanced by gradient boosting are also argued to perform better than neural

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networks in some cases, particularly for skewed feature distributions and irregular data (McElfresh et al. 2023).

XGBoost has been a popular choice for deriving labels from *Gaia* low-resolution XP spectra (R = 20–100). This method has been used to determine $T_{\rm eff}$, $\log g$, and metallicities [M/H] for ∼200 million XP spectra on the scale of reference objects. For example, Andrae et al. (2023) used APOGEE as a reference set, augmented by very metal-poor objects from H. Li et al. (2022) in LAMOST. Using AllWISE photometry to help in stellar inference in dust-extincted regions, Rix et al. (2022) deploy XGBoost to infer stellar parameters from the XP spectra, using LAMOST as a reference scale to cover a broad range of stellar types. Khalatyan et al. (2024) transfer spectroscopic stellar labels to the ∼220 million *Gaia* DR3 XP stars using labels from the Bayesian tool `StarHorse` (Queiroz et al. 2023), taking reference objects from a number of the major surveys, in order to cover the full sky and the range of stellar types in *Gaia*. Their reference labels combine data from the large surveys of APOGEE, GALAH, LAMOST, RAVE, SEGUE and Gaia-ESO, as well as small samples of hot stars, subdwarfs, very metal-poor stars, and white dwarfs. XGBoost has been used to derive not only metallicities from low-resolution *Gaia* spectra but also α-enhancements for stars in low-extinction regions (Hattori 2025). It has been used to infer the ages for LAMOST (Wang et al. 2024), the ages for the APOGEE spectra (Anders et al. 2023) (also see Section 5.2.6), and to predict the stellar rotation periods from *Kepler* photometry (Gomes et al. 2024).

Convolutional neural network: a neural network that uses filters to extract patterns from data

Adversarial neural network: a neural network framework with a generator and discriminator that are trained in competition

5.2.5. Neural networks. Neural networks are computational frameworks inspired by the structure of neuron connections in the brain (Rosenblatt 1958). They are built with layers of nodes that learn patterns embedded in the data by adjusting weights that determine how much influence each input has on the final output. They offer a very flexible modeling choice and have been used extensively for inferring labels from stellar spectra in data-driven modeling applications.

Different data-driven modeling frameworks that use neural networks have been widely applied to low-resolution *Gaia* XP spectra using external reference sets (e.g., Zhang et al. 2023, J. Li et al. 2024, Martin et al. 2024) to infer stellar parameters as well as α-enhancements (Laroche & Speagle 2025), as well as [C/Fe] from metal-poor spectra (Ardern-Arentsen et al. 2025) and stars with high [N/O] and [Al/Fe] abundances (Kane et al. 2025). Connecting the information in multiple spectroscopic and photometric surveys, Green et al. (2021) use a neural network model to map stellar parameters to broadband photometry. Similarly, Duque-Arribas et al. (2025) determine M-dwarf metallicities from photometry given a spectroscopically labeled training set. Data-driven neural network models have also been used on *Kepler* light curves to infer stellar parameters, rotation periods, and asteroseismic parameters (Blancato et al. 2022).

At higher resolution, APOGEE-NET (Olney et al. 2020, Sprague et al. 2022) and BOSS-NET (Sizemore et al. 2024) use discriminative convolutional neural network models to test correlations of information in spectral features that predict stellar parameters from APOGEE and BOSS spectroscopic data. The `StarNet` convolutional neural network of Fabbro et al. (2018) predicts APOGEE stellar parameters from APOGEE spectra. Using a generative adversarial neural network modeling approach, with their `StellarGAN` framework, Liu et al. (2024) classify both BOSS and APOGEE spectra. The discriminative neural network modeling approach of ASTRO-NN developed by Leung et al. (2023) delivers individual element abundances using censored portions of the spectrum and allows incomplete (including missing labels) and noisy training data. ASTRO-NN has delivered spectroscopic distances and ages for the APOGEE survey. Stone-Martinez et al. (2025) implement a probabilistic generative neural network approach with normalizing flows to similarly deliver ages as well as masses and distances from APOGEE spectra. A conditional invertible neural network model (with a reversible mapping) has also been used with spectra from

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the GIRAFFE instrument (HR10 and HR21 setups; $R = 20{,}000$) (Candebat et al. 2024) to infer stellar parameters and their uncertainties. Furthermore, a convolutional network employed by Ambrosch et al. (2023) to predict stellar parameters and a subset of abundances (Al, Mg, and Fe) for Gaia-ESOGIRAFFE spectra, using labels derived from physical models for the reference objects, with a view toward upcoming 4MOST and WEAVE survey spectra.

5.2.6. Low-resolution spectra. Data-driven methods are particularly valuable at low resolution because in this regime blended and asymmetrical features render the classical approach to measuring abundances—using individual lines in narrow regions that meet the isolation criterion—unfeasible (Wheeler et al. 2020, Kordopatis et al. 2023). Data-driven models have been very successful in accessing information encoded in low-resolution spectra that have not been previously extracted, including individual abundances and ages. Generative data-driven methods that include flux uncertainties also propagate well to low SNRs.

Individual elements, including C, N, O, Mg, Al, Si, Ca, Sc, Ti, Cr, Mn, Mn, Fe, Co, Ni, Cu, Ba, and Eu, have been inferred from millions of $R = 1{,}800$ and $R = 5{,}000$ spectra, along with ages (e.g., Ho et al. 2017a,b; Xiang et al. 2019; Wheeler et al. 2020; Z. Li et al. 2022; Zhang et al. 2024).[6] The data-driven metallicities inferred from millions of $R = 20$–110 *Gaia* spectra have revealed the "metal-poor 'heart'" (Rix et al. 2022, p. 45) and "metal-rich knot" (Rix et al. 2024) of the Milky Way, as well as the three-phase evolution of the Milky Way seen in the orbit–metallicity plane (Andrae et al. 2023, Chandra et al. 2024). Data-driven methods are also effective for deriving stellar parameters and individual abundances from $R = 1{,}800$ resolved spectra from the Multi Unit Spectroscopic Explorer Integral Field Unit (MUSE IFU)—including in extragalactic systems (Wang et al. 2022, Ness et al. 2025). This is achieved using a model trained not on MUSE spectra, which lack the necessary reference labels, but on Milky Way spectra observed at the same resolution and over the same wavelength range, using a different instrument: LAMOST spectra labeled with GALAH or APOGEE abundances.

There is a risk associated with element inference at low resolution using data-driven models that use the full spectral region, as these models may leverage correlations in the data to learn labels. If the correlation structure among abundances differs between the training and test sets, the inferred abundances may be inaccurate. This can be mitigated using generative modeling, with the ability to reconstruct the spectra given inferred labels, and with physical priors (e.g., `Data-Driven Payne`). Nevertheless, even for individual elements, it is often physically justified to use the full spectrum, as the relevant information may be distributed across a wide wavelength range.

Lowe et al. (2025) show that there are nontrivial differences in scale of the stellar parameters between their stellar parameter labels, which are determined using $R = 10{,}000$ spectra, and those of data-driven modeling of the *Gaia* low-resolution spectra. The large differences (and lack of correlations) for the same stars seen between some of the catalogs suggest either large discrepancies between the reference object scales in the automated data-driven modeling approaches used or bias in the test-time results for some methods in some parameter spaces. This highlights a key challenge in data-driven label transfer, which is validating the model's performance beyond the reference objects used for training.[7] Failures or biases can arise from extrapolation at test time

[6]The analysis by Ting et al. (2017) documents the expectation of precision as a function of resolution and SNR for different surveys. Even at a lower resolution than that of DESI, (R = 5,000), the precision that can be obtained in principle for up to a dozen abundances of elements is reasonably high, at 0.02–0.1 dex (5–25%). However, at R = 2,000, uncertainties remain prohibitively great for most of the neutron capture elements.

[7]It is possible that a data-driven model shows good performance in cross-validation but fails at test time in some instances.

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beyond the training domain, substantial differences in spectral quality between training and test data, or signals such as sky and telluric residuals or diffuse interstellar bands in the spectra [see Ho et al. (2017b), who masked diffuse interstellar bands to improve label accuracy]. Overlapping stars between surveys provide a critical opportunity to identify regions of disagreement. In this context, generative models offer an advantage, as they allow one to assess the consistency between model-generated and observed spectra given the inferred labels.

5.2.7. Evolutionary state from stellar spectra via learned asteroseismic parameters. Data-driven modeling has enabled information to be extracted from stellar spectra pertaining to evolutionary state via the asteroseismic parameters. Although stellar parameters of $T_{\rm eff}$ and $\log g$ alone are discriminative of red clump and red giant stars (e.g., see Bovy et al. 2014) due to mixing and dredge-up dependency associated with the occurence of the helium-flash, there should be additional differences in C and N abundances that can be detected in spectra (Martell 2008, Masseron & Gilmore 2015, Masseron et al. 2017, Hawkins et al. 2018). These signals can be used to automate the classification of evolutionary state from stellar spectra. Hawkins et al. (2018) subsequently deployed `The Cannon` in order to learn asteroseismic parameters from the spectra (also see Ting et al. 2018b, Lu et al. 2022). They inferred from APOGEE spectra both the mode period spacing ($\Delta\Pi$) and large frequency separation (Δ_ν) (see **Figure 5**). The reference set labels were taken from Vrard et al. (2016), determined from asteroseismic measurements of the multiepoch *Kepler* photometry.

Data-driven models of the spectra have also been used to identify features that are associated with evolutionary state differences. Banks et al. (2023) use `The Cannon` and XSHOOTER spectra to search for the signatures of red giant versus red clump stars. They used the interpretability of the simple polynomial model, in which the first-order coefficients identify which wavelengths are informative for assigned labels. They identify CN, CO, and CH features as the primary sources of evolutionary state information and document 215 informative features across the 0.22–2.5-μm spectral range. Banks et al. (2024) extended this work to use Veloce Rosso spectra to confirm molecular features with carbon as the primary sources of information to discriminate

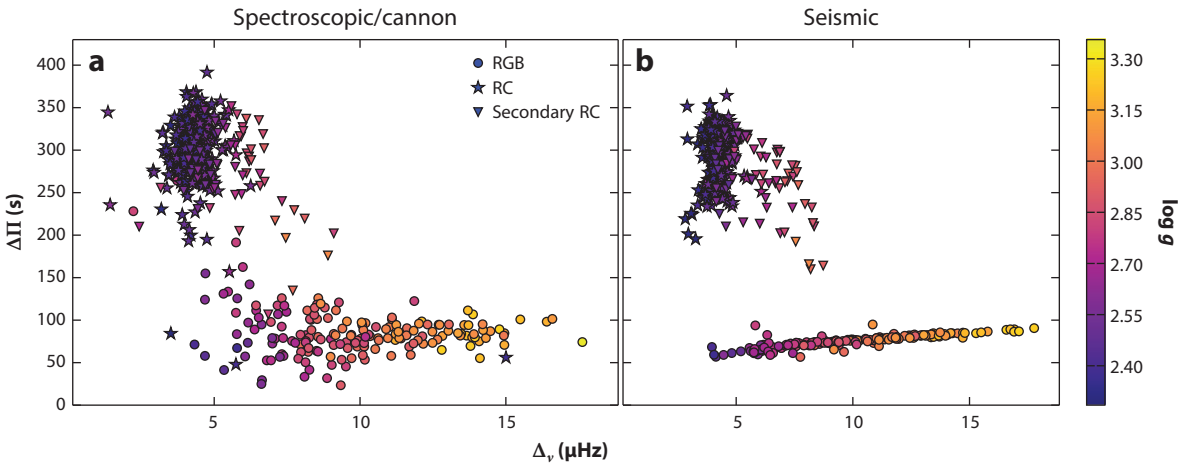


Figure 5

The separation of spectra into evolutionary state by inference of period spacing ($\Delta\Pi$), and large frequency separation (Δ_ν), with labels inferred from the spectra with `The Cannon` (*left*) and labels measured from asteroseismology (*right*). Figure adapted from Hawkins et al. (2018) with permission of the AAS. Abbreviations: RC, red clump; RGB, red giant branch.

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evolutionary states and found differences in isotopes of carbon (^{12}C, ^{13}C) between red clump and red giant stars.

5.2.8. Stellar ages and distances for red giants with label transfer. Data-driven modeling has opened up the reporting of spectroscopic ages for red giant stars and, therefore, enabled large-scale age mapping across the Milky Way. Conventionally, stellar ages are determined using photometry and isochrones or asteroseismology. Using measurements of absolute magnitude and color, stellar ages can be determined with relatively high precision (<10%) for stars at the turnoff and subgiant evolutionary phase, where there is discriminative power of stellar isochrones. *Gaia* has been a transformative contributor in this domain (e.g., see Xiang & Rix 2022). Asteroseismic missions have similarly revolutionized the field by their samples of stars with precision ages (<10%) (Borucki et al. 2010, Ricker et al. 2014). The APOKASC sample of ∼10,000 stars (Pinsonneault et al. 2018, 2025) observed by both *Kepler* and APOGEE has particularly opened up new avenues in galactic archaeology. Reference stars with ages in the large spectroscopic surveys enable ages to be propagated to other stars in the survey using the spectra directly, although at relatively low precisions of 20–30%. This is achieved by leveraging correlations from either abundances with ages due to galactic evolution (e.g., chemical clocks) or stellar evolution, i.e., carbon and nitrogen features in the red giant spectra that inform age, due to mass-dependent dredge up (Masseron & Gilmore 2015, Martig et al. 2016). See the sidebar titled Age Estimates for a summary of when different age dating methods are applied.

The first data-driven age maps derived using spectra were reported by Ness et al. (2016), Martig et al. (2016), and Ho et al. (2017a) for red giant stars, using both the spectra directly and

AGE ESTIMATES

Individual stellar ages are estimated using a variety of methods and rely on models (see Soderblom 2010 for a review). Different methods are applicable in different evolutionary regimes.

Isochrone Ages

Isochrone ages are most precise for turnoff stars, where there is discriminating power offered by absolute magnitude and color (or equivalently, from $T_{\rm eff}$ and log g) and stellar isochrones separate at the turnoff in age at fixed metallicity. Coeval stellar populations, such as globular clusters, can have their ages estimated via isochrone fitting to the full population.

Asteroseismic Ages

Asteroseismic ages are determined for red giant stars via measuring stellar mass through internal oscillation frequencies that is converted to age via stellar models given a star's metallicity using brightness variations measured for stars from the asteroseismic missions like *Kepler* and $T_{\rm eff}$.

Spectroscopic Ages

Spectroscopic ages have been enabled by data-driven methods, and inferred from stellar spectra by leveraging information in the spectrum that is correlated with age. Learning these correlations relies on a reference set of stars with both age measurements, e.g., from asteroseismology or isochrones, and observed stellar spectra for those stars. Data-driven methods that infer stellar labels, such as in the case for spectroscopic ages, propagate the scale of the reference objects. Subsequently, the accuracy of the labels is always limited by that of the reference objects. In the case of age estimates, these are model-dependent and can vary substantially under different assumptions (e.g., Morales et al. 2025).

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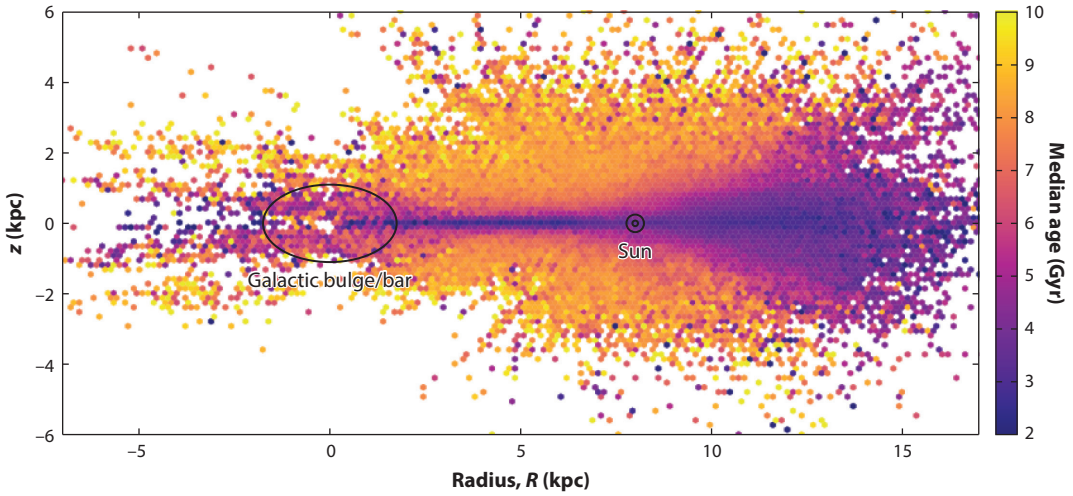


Figure 6

The age map of ∼400,000 red giant stars in the Sloan Digital Sky Survey V's Galactic Genesis survey generated using normalizing flows using a reference set of stars from APOKASC (Pinsonneault et al. 2025). This shows the signature of the Milky Way's inside-out formation, with older stars in the inner galaxy and younger stars in the outer. Figure adapted with permission from Stone-Martinez et al. (2025) (CC BY 4.0).

carbon and nitrogen measurements. The reference labels for these were taken from the APOKASC sample. Large age maps—from inferred masses of red giant stars that can be transformed to ages using stellar models given a known metallicity—for hundreds of thousands of stars are now available using data-driven label transfer. **Figure 6** shows the age map derived using a normalized flow model and the APOKASC reference labels (Stone-Martinez et al. 2025). Several data-driven methods for inferring stellar ages also include spectroscopic distances, which are inferred in the same framework (e.g., Leung & Bovy 2019, Eilers et al. 2022a).

Data-driven ages have served large-scale mapping pursuits to pin-point signatures of evolutionary processes in the Galaxy. Spectroscopic ages have been used as inputs to models to learn that radial migration is substantial in the Milky Way (Frankel et al. 2018). Ages of stars in the bulge region inferred using `The Cannon` have shown that high-α disk stars are old at all Galactic radii, and the age gradient comes largely from the low-α. Similar to studies of microlensed dwarfs (Bensby et al. 2017), spectroscopic ages include a fraction of younger stars (age ∼2–5 Gyr) at high metallicities (Hasselquist et al. 2020, Sit & Ness 2020). The age–abundance relations across the disk examined by Yuxi et al. (2022) show that the gradients in age and element abundances vary across metallicity, [Fe/H], and α-enhancement. These age–abundance relations capture important signatures that may be set by the galactic bar (Zhang et al. 2025).

5.2.9. Labeling M dwarfs using novel data-driven approaches. M-dwarf analysis is particularly challenging; the spectra of cool stars ($T_{\rm eff} < 4{,}200$ K) are densely populated with molecular lines, and there are uncertainties in the quantities describing atomic and molecular physics. This renders conventional methods of using equivalent widths of absorption features or spectral synthesis of selected lines prohibitive. Many absorption features are unidentified or have inaccurate reported oscillator strengths (see Henry & Jao 2024 for a review, as well as Hoeijmakers et al. 2015, McKemmish et al. 2019, Ishikawa et al. 2020, Souto et al. 2022, Jahandar et al. 2025). Using a high-fidelity reference object, Jahandar et al. (2024) identify thousands of absorption lines in the spectra that are not present in the models, and vice versa. Using partial information and small

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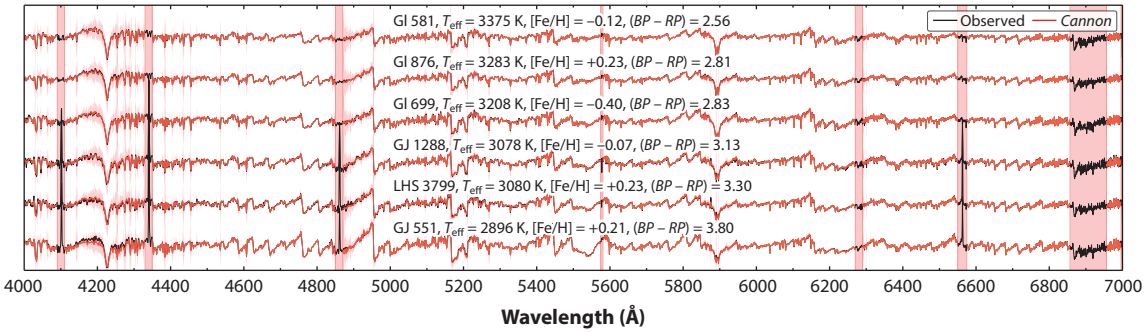


Figure 7

An example of M-dwarf spectra and their data-driven spectral models generated at the best-fit parameters; these well fit the observed spectra. `The Cannon`'s model (*red*) and observed spectrum (*black*) are shown for six stars observed by WiFeS at $R = 3{,}000/7{,}000$. The vertical red bars correspond to regions that were masked. Figure provided by and adapted with permission from Rains et al. (2024) (CC BY 4.0).

ensembles of carefully examined stars Jahandar et al. (2024) pinpoint a short list of a few hundred lines out of a total of 10^4 that are reliable to infer properties of M-dwarf spectra, including $T_{\rm eff}$, $\log g$, [Fe/H], and 15 individual abundances.

Data-driven modeling offers an effective analysis route for these stars–which comprise the majority of the stellar population. In this realm, data-driven models have the ability to extract information that would otherwise remain inaccessible. Alongside data-driven models, innovative approaches to obtaining reference object labels are required. A successful novel workaround to obtaining reference labels has been to exploit M dwarfs in binary systems to construct the reference set. For M dwarfs in binary systems, the (hotter) companions, at temperatures in which conventional abundance analysis is more accurate ($T_{\rm eff} > 4{,}500$–$6{,}500$ K), can be used to label the M-dwarf stars with metallicity and abundances under limiting assumptions. Adopting the labels of the companion star is based on the premise that binaries have shared abundance signatures, but we note that this assumption is sometimes violated (Oh et al. 2018, Yong et al. 2023). Correlations between stellar abundances and Galactic kinematics are also used to inform reference star abundance labels (Rains et al. 2024). Several studies have inherited the binary star labels for a hotter-temperature star with reliable measurements from physical models (Maldonado et al. 2020, Rains et al. 2024, Qiu et al. 2024, Behmard et al. 2025). Models that use the binary companion as the reference label abundances have been implemented using both `The Cannon` and support vector regression. In this way, a model can be built from the reference objects and propagated to other M-dwarf stars without requiring the M-dwarf abundances come from the limited ab initio models. **Figure 7** shows examples of M-dwarf spectra and generated model spectra using `The Cannon` with WiFeS optical (R = 3,000/7,000) spectra.

5.3. Data-Driven Label Calibration

To achieve precise and accurate stellar parameters and abundances, survey teams undertake considerable efforts toward self-calibration and homogenization of measurements derived from the spectra (e.g., Smiljanic et al. 2014; Holtzman et al. 2015, 2018; Nidever et al. 2015; Jofré et al. 2017; Pancino et al. 2017; Jönsson et al. 2020; Abdurro'uf et al. 2022; Kos et al. 2025; Meszeros et al. 2026). This procedure can include using reference objects like open and globular clusters to model and subtract temperature trends at fixed overall metallicity, and shift absolute scales to the Solar Neighborhood reference. There are typically still residual trends between individual

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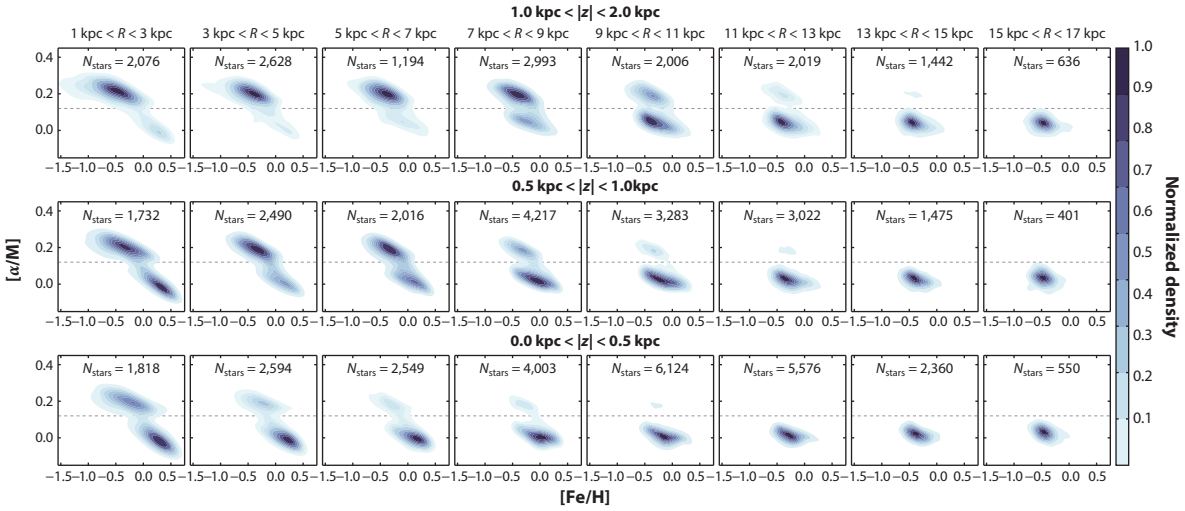


Figure 8

The [Fe/H]–[α/M] plane across Galactic radius made from calibrated abundance data whereby the systematic residual trends across evolutionary state have been removed, shifting all stars to the abundances they would have at $\log g = 2.2$. This removes nontrivial systematics across $T_{\rm eff}$ and $\log g$ in the element abundances and leads to a more distinct separation of the high- and low-α sequences in the spatial plane. Such corrections as implemented by Eilers et al. (2022a) are critical for the spatial mapping pursuits enabled by large surveys. Figure adapted with permission from Eilers et al. (2022a) (CC BY 4.0).

abundances and stellar parameters that manifest when stars across evolutionary states are combined. This has typically meant that studies restrict themselves to narrow regions in the evolutionary state (e.g., Weinberg et al. 2019, Griffith et al. 2021, Ratcliffe et al. 2022, Lu et al. 2024) to avoid these systematics.

In work by Eilers et al. (2022a), a formal framework was developed to self-calibrate residual trends that persist in element abundances across evolutionary state. The systematics, whether real or inherited from inaccurate models, are not trivial. Eilers et al. construct a linear model that connects the individual abundances to both their orbital properties—upon which element abundances depend (e.g., Sanders & Binney 2015, Frankel et al. 2018, Ratcliffe et al. 2024)—and their surface gravity, which is a nuisance parameter to model and then subtract off. A fiducial value of $\log g = 2.2$ is used to effectively shift each star's abundance vector to effectively represent the abundance of a star at a particular evolutionary state. This calibration to a fixed evolutionary state means that scientific endeavor can take advantage of larger samples. Stars across large distances, which typically span a wide range in $T_{\rm eff}$ can then be examined in concert. The calibrated abundances are shown in the α-bimodality across radius, showcasing the superposition of two galactic disks in the Milky Way, as demonstrated in **Figure 8** (also see Nidever et al. 2014, Hayden et al. 2015, Queiroz et al. 2020, Imig et al. 2023). A data-driven calibration of the abundances is also implemented by Sit et al. (2024).

5.4. Residuals in the Spectra Away From Data-Driven Models

Data-driven models serve as tools for discovery in both spectra and abundances. The residuals of observed spectra, subtracted from the spectra generated by the model given the inferred labels, make data-driven modeling an efficient tool of variance analysis and outlier identification on a per-wavelength basis. Residual assessment from data-driven modeling can both quantify the reliability

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of labels that generate the spectra and tap into the physics underlying the formation and evolution of stars, as well as pinpoint the information that is not from the stellar photosphere.

For example, residuals from data-driven spectra modeling have been used to find activity and binarity in the Gaia RVS spectra (e.g., Angelo et al. 2024). Outlier identification is achieved directly in the spectral domain. Wheeler et al. (2021) implement a local linear modeling projection to learn the behavior of the neighborhood in the spectrum space for this purpose. Although originally in pursuit of chemical tagging—the goal of tagging stars to their birth sites given their abundance vectors—using the spectra directly, McKinnon et al. (2024) determined that the simple five-label model ($T_{\rm eff}$, $\log g$, [Fe/H], [α/Fe], and age) that they fit to the APOGEE flux provides a remarkable fit to the spectra. For the vast majority of wavelengths, their model reproduces the spectra within uncertainties. Residual deviations instead derive from velocity-dependent features associated with sky signals and the interstellar medium. Therefore, approaches to improve sky subtraction will help improve measurement precision, and data-driven models of the spectra can lead to discovery and analysis of diffuse interstellar band signals (also see Saydjari et al. 2023). In addition, spectral decomposition can significantly improve the precision of RV measurements (Saydjari et al. 2025). Higher-precision RVs are also achieved using a template model built from the data itself instead of a best-fit ab initio model (Nidever 2023). Interpretable data-driven methods that quantify the origin of the information can also be used for line discovery. Careful analyses by Cunha et al. (2017) and Hasselquist et al. (2016) identified Ce and Nd spectral features in the APOGEE spectra. Approaches inspired by these successes could be automated in data-driven models.

5.5. Data-Driven Nucleosynthesis and Dynamics

Data-driven modeling of stellar element abundances has revealed that the chemical abundance space of the Milky Way disk is lower in dimensionality than previously expected. This finding is surprising given that an assumption underpinning galactic archaeology in the past decades has been that the disk abundance space would be high dimensional, reflecting diverse nucleosynthetic pathways. The restricted abundance dimensionality of disk stars makes pursuits such as chemical tagging (Freeman & Bland-Hawthorn 2002), which has been a motivation of multiple large surveys in which stars are tagged to their siblings using available abundances, prohibitive, even at precisions of 5% (Ness et al. 2018a, Casamiquela et al. 2021, Manea et al. 2024, Sinha et al. 2024). However, residuals away from data-driven models are potentially critically informative on the birth radius and potential birth sites if the measurement precision is sufficiently high (e.g., Ness et al. 2022, Ting & Weinberg 2022, Weinberg et al. 2022, Griffith et al. 2024).

Remarkably, the precision with which the supernova elements of O, Al, Si, Ca, Ti, Cr, Mn, and Ni can be predicted by two elements alone (Fe, Mg) with a simple local linear regression[8] for stars over a range of present-day galactic radius and metallicity values is only marginally lower than that of stars in open clusters (Ness et al. 2022, Sinha et al. 2024). The precision of the prediction represents the homogeneity of the Milky Way disk in supernova elements at fixed age and galactic birth radius, which is on the order of ∼5% (0.02 dex). This homogeneity is similar to that of external galaxies over 600-pc scales (Kreckel et al. 2019). Although abundances do not tag cluster birth sites even at 5% precisions, they can tag birth radius and age (Bedell et al. 2018, Ness 2018, Casali et al. 2020, Hayden et al. 2022, Casamiquela et al. 2021, Yuxi et al. 2022, Molero et al. 2023, Ratcliffe et al. 2024, Shejeelammal et al. 2024), and the neutron capture elements are particularly

[8]A local model is constructed separately for each data point, where the target label is inferred from its neighborhood. Specifically, a straight line is fit to the nearby data by minimizing the sum of squared differences between observed and predicted values. Each point therefore has its own linear model, defined by a slope and an intercept.

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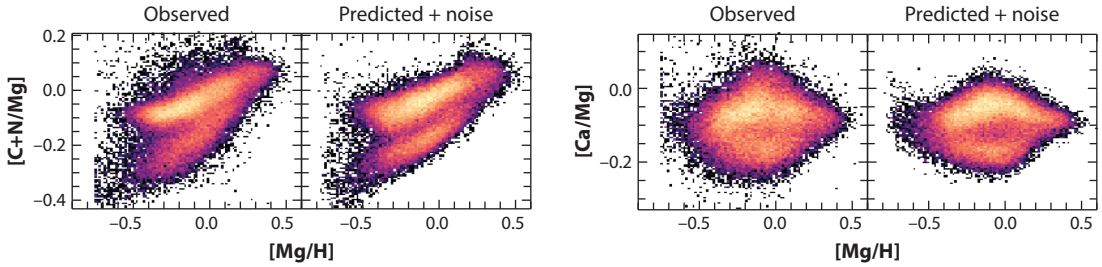


Figure 9

The generated elements from a model of only $k = 2$ underlying processes. Left halves of the panels show the observed abundance distributions, and right halves of the panels show the predictions, including noise. Figure adapted with permission from Griffith et al. (2024) (CC BY 4.0).

powerful along these axes (Bedell et al. 2018, Spina et al. 2018, Tautvaišienė et al. 2021, Casali et al. 2023, Manea et al. 2024, Mead et al. 2025).

Data-driven models from Griffith et al. (2024) of ∼20 element abundances in APOGEE show that these abundances lie on a low-dimensional manifold and can be well represented by $k = 2$ or 3 underlying processes. These processes generate the observed abundances through combinations of relative source contribution, as shown in **Figure 9** (also see Weinberg et al. 2019, 2022). Intrinsic (residual) information is present at the 1–10% level and shows spatial correlations (Griffith et al. 2025). This intrinsic information and correlations between residual abundances contains information related to the stars' nucleosynthetic history (Ting & Weinberg 2022).

Using a simple local linear regression of the data, Mead et al. (2025) find that 31 element abundances of the disk (C, O, Na, Mg, Al, Si, P, S, Ca, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Sr, Y, Zr, Ba, La, Ce, Pr, Nd, Sm, Eu, Gd, and Dy) are rigidly coupled, as shown in **Figure 10**. They demonstrate that the iron peak, light, and alpha elements can be predicted to within 5% in the disk by (Fe, Mg) alone and the neutron capture elements to within 10% using (Fe, Mg) at solar metallicity. A four-element predictive model (Fe, Mg, Y, Ce)—that includes first- and second-peak *s*-process elements as predictors—improves the neutron capture element predictions to be within 5% of their measured values. These models, that report the intrinsic scatter of the model predictions, enable the calculation of the *C*-space of large surveys. The *C*-space is far in deficit of what is required for chemical tagging. Similar to Griffith et al. (2025), who explored the residuals of model prediction spatially, Mead et al. (2025) report residual correlation structure, but in the interelement residual correlations. As highlighted by Ting & Weinberg (2022), this illustrates that key information is encoded in the relationships between the residual abundances. The spatial variations and residual correlations are directly related to the origin of the elements and how they are transported in the evolution of the Milky Way Galaxy. They provide key constraints for chemical evolution models. See the sidebar titled High-Fidelity Abundance Measurements for a description of spectroscopic goals that data-driven models have both established and serve.

Tests using the spectra also show underlying correlation between abundances (see Feeney et al. 2021, de Mijolla & Ness 2022). Principal component and latent-variable models describe the abundances or spectra with ≥6 components of independent information (Ting et al. 2012, Price-Jones & Bovy 2018, Casey et al. 2019). These frameworks capture both the dominant generative sources and the smaller-amplitude residual structure measured from generative 2–4-source models. However, some of these components may be attributable to nonstellar signals, or model systematics and noise (e.g., Saydjari et al. 2023, McKinnon et al. 2024).

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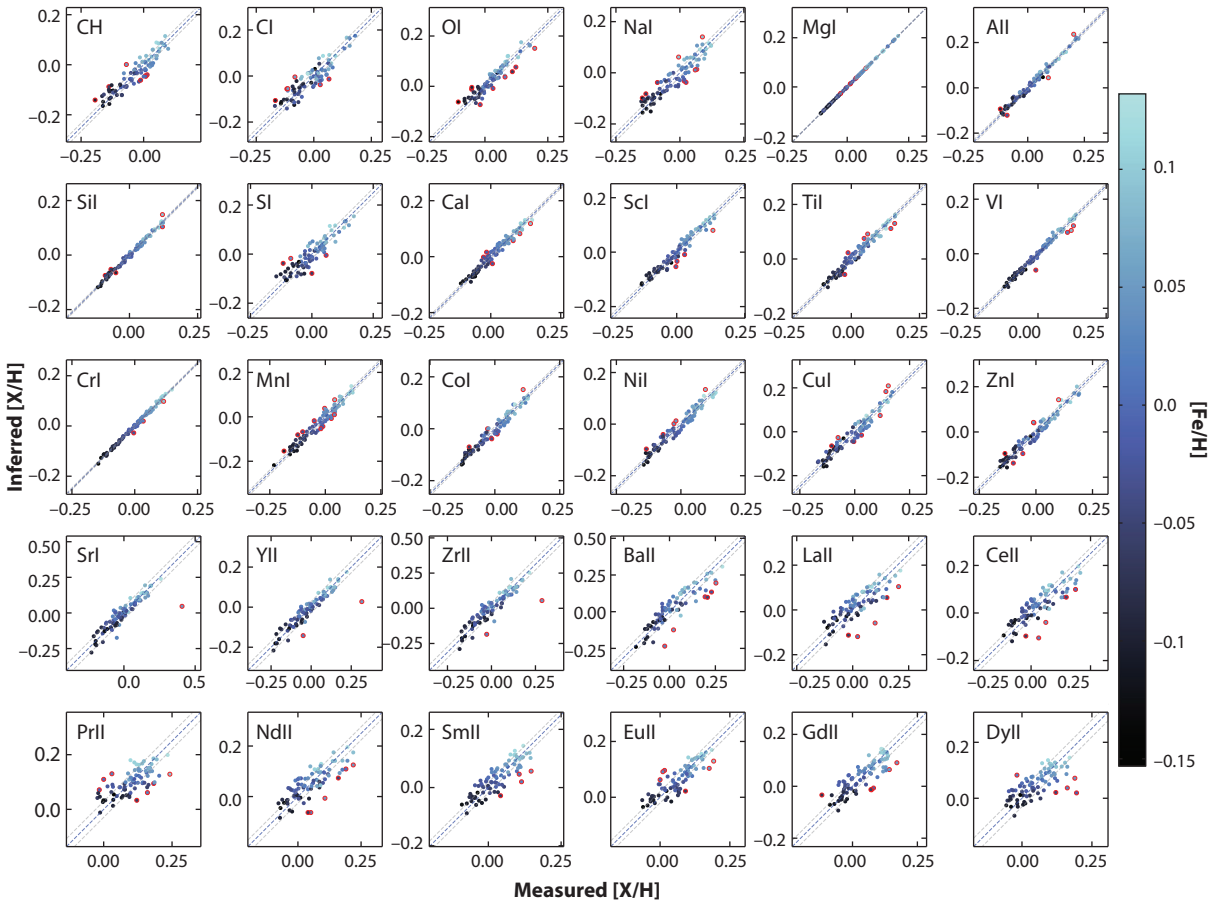


Figure 10

The measured versus inferred abundances using a regression of $T_{\rm eff}$, log g, [Fe/H], and [Mg/Fe], showing that 30 elements are predicted by 2, to within 2–10% at [Fe/H] ∼ 0. This precise inference persists across a wider metallicity range (see Mead et al. 2025). This data-driven analysis demonstrates that the elements of the disk are rigidly coupled, and precision goals of <5% (for >30 elements) are required by next-generation disk surveys to access individual element information. Figure adapted with permission from Mead et al. (2025) (CC BY 4.0).

Data-driven dynamics, as introduced by Price-Whelan et al. (2025) has demonstrated that abundances can be used to infer properties of the Galactic mass distribution directly (see Horta et al. 2024b). This approach circumvents the need to account for complex selection functions and completeness corrections. In particular, residuals from model predictions may provide valuable insights for data-driven dynamics (Frankel et al. 2025).

6. LIMITATIONS OF DATA-DRIVEN MODELS

Data-driven approaches applied to spectra and photometry for label transfer are typically fast and make spectroscopy accessible. Therefore, data-driven modeling can vastly decrease the processing time of surveys and support flexibility in the choice of reference objects and label scale to propagate to larger datasets. Importantly, data-driven models will inherit any biases and inaccuracies of the training data and propagate this to new data. They are therefore, at this stage, typically

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HIGH-FIDELITY ABUNDANCE MEASUREMENTS

Abundances measured to precisions of 2% are available for the highest SNR subsets of large survey data for a subset of elements and for small high-fidelity samples of stars observed at high resolution (R = 60,000–110,000) (e.g., Meléndez et al. 2010; Adibekyan et al. 2012; Yong et al. 2013, 2023; Bensby et al. 2014; Bedell et al. 2018; Nissen & Gustafsson 2018; Spina et al. 2018; Adibekyan et al. 2020; Casali et al. 2020; Tautvaišienė et al. 2021). Despite their small sample sizes of 100–1,000 stars, the small boutique samples serve as critical benchmark element abundances for the Milky Way. Measurement uncertainties for 30 elements of 2–5% represent an important goal for future large surveys (10^{5-6} stars). This goal supports a substantial amount of scientific endeavor, including the emerging pursuit of data-driven nucleosynthesis and residual abundance chemical evolution modeling. When large surveys can reach the quality of the current boutique samples of high-resolution Milky Way spectra, this will be paradigm-shifting in our understanding of the Galaxy over a new range in scale, directly connecting elemental origins to Galactic origins. Data-driven analyses have a role to play in both precision label inference and subsequent population analysis of abundances and dynamics.

implemented as tools of precision and not of accuracy; and these serve different scientific goals (see Nissen 2015). There remains an important opportunity to use data-driven methods in the pursuit of accuracy, as a mechanism of feedback and constraints in theoretical modeling frameworks and inputs. To pursue accuracy and work toward more complete physical representations of stars, detailed analysis of small high-quality benchmark samples, as well as continued development in the realm of more complete physical models, is key (e.g., Masseron et al. 2021, Aerts & Tkachenko 2024, Lind & Amarsi 2024, Reyes et al. 2025).

Data-driven models are only premised in a ground truth where the test data are representative of the training. In label-transfer applications, if the reference objects used to train a model are not fully representative of the entire survey parameter space that the model is deployed on, the full ensemble of survey stars may not be properly labeled, i.e., in the parameter space whereby the model is extrapolating. In using data-driven models as a tool for labeling spectra therefore, the reference objects used to build the model should be representative of the test objects in which the labels are tied to a physical basis. This is not always trivial to ensure in high-dimensional label spaces, although generative models can test the fidelity of the inferred labels. Spectra used at test time should also be representative of the training, in terms of reduction, calibration, and normalization. If a subset of the test spectra contains nonstellar signals like diffuse interstellar bands or sky features not present in the reference objects, this may also interfere with the accuracy of the inferred labels, although this is not a problem specific to the data-driven framework. It has been demonstrated that data-driven models do perform well when trained on one instrument and applied to data from another. This is presumably because signals from the instrument or reduction in training time do not correlate with the labels, so such signals may inflate the model uncertainty but do not impact or bias the label inference.

Data-driven models can, in principle, produce incorrect labels if they rely on correlations in the spectra rather than on spectral features that are intrinsically tied to the label. For example, a flexible model trained on the full wavelength range may learn to predict the abundance of an element that is not directly observable in the spectrum, simply by leveraging correlations with other elements or stellar parameters. In some cases, this is physically motivated—oxygen, for instance, influences broad regions of the spectrum through molecular equilibrium and opacity effects (e.g., Ting et al. 2018a). However, if such correlations differ between the training and test sets, the resulting predictions can become biased. This highlights the need to distinguish correlation from

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causation in label inference. Comparison of the outputs of different techniques used to infer labels for the same stars is one way to identify problems and assess the impact of correlations versus causation. Evaluating the goodness of fit of the model generated spectrum compared to the observed spectrum is another.

Data-driven models can also lack physical interpretability. Data-driven automated machine learning methods work, but it can be unclear why and from where the information is derived. For example, though flexible models like neutral networks can very successfully recover stellar labels at high precision, they may also provide no insight as to how this taps into the information encoded in the spectrum in any physical way. This can be mitigated by using simple and interpretable models (i.e., models that can be queried to understand how inputs and outputs are connected), such as a linear form. However, simple interpretable models like a first-order polynomial are also the least flexible. Local-linear models may offer a good path forward, allowing both flexibility and interpretability, although at more computational expense than single models. Another approach is to constrain models to a simple form in a latent space, preserving interpretability while capturing the structural properties of the relationship between data and labels.

Ting (2025) provides a thorough review on the foundational limitations of machine learning regression. By design, the regression procedure compresses the true values, thereby "systematically underestimat[ing] extreme values." This is often seen in the cross-validation of labels at the extreme ends of the range. This is reported by Ting (2025) to be a relevant effect when the ratio of the intrinsic signal range to the measurement uncertainty is <10, which is often the case in astronomy. This is a good example of why data-driven methods and modeling have utility but are not a replacement for automation using physics-based ab initio models. However, the attenuation bias (see Ting 2025, p. 2) behaves in a predictable way, informed by the statistical properties of the spectral features and scaling with SNR, potentially allowing data-driven calibration of this effect. Ting (2025) suggests that full spectral fitting including individual wavelength values may help reduce the impact of these biases (rather than the use of indices or summary statistics as data inputs to the model) and restricted models like a linear regression or full Bayesian hierarchical models may allow for easier calibration and reduced overall bias compared to alternatives. Ting (2025) also cautions against "error-ignorant" frameworks such as discriminative models, which serve as mapping functions from spectra to labels and treat noise in the data as informative features. Generative models are typically better suited for propagating the model learned on high-SNR high-fidelity data to low-SNR or lower-resolution data, assuming there is an implicit or explicit inclusion of propagated uncertainty, whether in the model as a variance or terms in the likelihood function directly.

7. FUTURE PROSPECTS AND CONCLUSION

Data-driven analysis has played an important role in the growth and evolution of observational astronomy. The utility and precision of data-driven methods and their relatively modest computational cost make them useful, accessible, and convenient. Large-scale surveys, the ongoing increase in computing power, and the development of modern computational infrastructure will continue to transform astronomy and drive development in this area. Data-driven models built for cleaning, calibration, and component separation of spectra will be important across redshift domains, from studies of stellar ecosystems to galaxies.

For the in situ stars of the Milky Way disk, a salient challenge has emerged from data-driven modeling, which is the need for very high abundance precision of sub-2–5% (or <0.01–0.02 dex) for >30 elements across light, iron-peak, and α-elements, as well as neutron capture element

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families, in order to access their independent information and maximize survey *C*-space.[9] In the disk, where elements are rigidly coupled, precision is key, and data with these uncertainties have the capability to provide a readout of nucleosynthesis and access the variability in the parameters of the star formation history, which is encoded in it. With such demands on the fidelity of the abundance precision, stellar astrophysics is not dissimilar from exoplanet pursuits, which require similarly high-fidelity measurements propagated from high-quality spectra. Meeting these abundance precision demands in large surveys will revolutionize the scale at which we understand the origin and redistribution of the elements over cosmic time across the Galaxy in stellar ecosystems. To achieve high precision at scale, from SNR > 100 per pixel spectra in the optical (R > 20,000), data-driven models will likely play a central role, including the processing, cleaning, and decomposition spectra to remove spurious signals, for label inference, label calibration, and updating and correcting the physics underlying theoretically derived physical models. Future large volumes of spectra of high quality (R > 100,000, SNR > 100) also facilitate the transition into the data-driven isotopic era of elemental abundances (e.g., with Mg isotopes; see McKenzie et al. 2024).

There is a substantial volume of new low-resolution data of the Milky Way halo, along with integral field unit data from other galaxies that will be delivered in the coming decades. These data, combined with data-driven modeling techniques, will support an era of extragalactic archaeology. It will be possible to leverage data-driven methods to compare and contrast the Milky Way and other stellar populations, via placing them directly on a consistent element abundance scale. The higher element abundance diversity of the Milky Way halo compared to the disk also means that element abundance precisions of around 25% or <0.1 dex in the halo appear to have substantial value for chemical tagging of the halo to progenitor galaxy analogs. High-quality spectra of dwarf galaxies obtained with ≥8-m class telescopes offer a critical validation of the results and conclusions drawn from the data-driven modeling. Data-driven models for the purpose of element abundance inference have been applied only to resolved stars. However, they could also be extended to analysis of unresolved stellar populations—empirical libraries already exist to support this.

Although data-driven methods have very organic applications in the large-data realm, including for parameter and abundance homogenization, spatial mapping of abundances, outlier identification, and in connecting the Galactic and extragalactic domains, in terms of labels, they ultimately depend on the quality of the training set. These methods are only as accurate as the best theoretical models and benchmark stars with fundamentally determined parameters (e.g., interferometric $T_{\rm eff}$ and stellar radii, and stars with dynamical masses). In contexts in which information in spectra is connected to physical labels, data-driven models are not a replacement for theoretical models. Rather, they are a tool to use alongside them. Progress in the data-driven landscape therefore relies on the continued efforts toward building models from first principles that are more physically complete representations of stars. Learning where, why, and how stellar models built from the data diverge from models built from first principles offers information to correct ab initio representations. Deconstructing how and why data-driven methods themselves work and identifying failure modes, focusing on small-scale signals and residuals away from the models, offer important insights for connecting to physical bases. Furthermore, experiments at different ends of spectroscopic approaches—from being fully driven and informed by theoretical models to a

[9]Introduced by Freeman & Bland-Hawthorn (2002), this represents the number of unique chemical cells spanned by a dataset. It quantifies the discriminating power of abundances: A larger *C*-space allows more detailed insights into the enrichment history of the disk and offers stronger constraints on chemical evolution models. Achieving a high *C*-space ($>10^6$) is one of the conditions for successful chemical tagging (Bland-Hawthorn et al. 2010). High-precision abundances are essential for maximizing the *C*-space of the data.

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fully data-directed approach—are useful. These efforts are complemented by and rely on the observation and study of benchmark samples of stars with high-quality spectra over a broad range in wavelength and photometry, observed by different instruments over multiple epochs.

As computing progress continues to advance the capabilities to liberally deploy machine learning and artificial intelligence, consideration and tests of the succession of inherited limitations and bias in output are critical. Several methods now exist that deliver labels for the same datasets, and different approaches enabling comparisons will serve as valuable validation. In particular, when results from different methods for the same objects diverge, it presents an opportunity to learn. Diversity in data, methodologies, and teams is essential to continue to drive progress toward developing a more complete understanding and interpretation of the information embedded in the multitude of observational data and to inform next-generation pursuits.

SUMMARY POINTS

1. Data-driven models directly enable large spectroscopic surveys to be placed on the same label scale.
2. These approaches have opened up new avenues and opportunities in Galactic studies including ages, evolutionary state and distances from stellar spectra, discovery of diffuse interstellar bands, and abundances from low-resolution spectra, at higher precision (although not necessarily higher accuracy), including at lower resolution and lower signal-to-noise ratios than previously.
3. An ensemble of methods exist that represent a trade-off among flexibility, domain use, computational expense, and interpretability. Models can inherit or impute biases, and different methods offer the means to compare results obtained under different formalisms. Incorporating uncertainties is important for ensuring reliable propagation to unseen data of varying quality.
4. Generative models are advantageous for label transfer because they allow verification that the inferred labels can generate the input spectra.
5. Data-driven models rely on theoretical models for many applications. In calibration applications they provide cleaner, betterized spectra and no theoretical framework is needed, but in labeling contexts they rest upon the fidelity of physical stellar models that set the reference scale.

FUTURE ISSUES

1. Data-driven models have revealed that the disk is rigidly coupled in element abundance space. Next-generation (large) spectroscopic surveys of the disk should aim to measure 30 abundances in the disk at precisions of sub-2–5% (<0.02 dex) to tap into intrinsic information in the elements. Precision requirements in the halo appear less stringent (<0.1 dex or 25%).
2. Data-driven models can serve extragalactic archaeology by placing stars in other galaxies on the same abundance scale as the Milky Way, and their utility can be extended to unresolved populations.

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3. Comparisons of results from the different methods for common stars offers a means of testing data-driven models and identifying limits of our understanding.
4. An important future opportunity is to use data-driven models to pursue accuracy, as tools to correct and update theoretical model inputs, and to update physical descriptions of stars and uncertain atomic and molecular data using information in the residuals of generated models compared to the data.
5. Data-driven spectroscopy can, and in the future hopefully will, link scientific domains. It derives methodological foundations in statistics and artificial intelligence and has applications in exoplanet and astrobiology studies and in stellar, galactic, and extragalactic archaeology, as well as broader physical modeling domains such as climate science.

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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